Unequal Agents:
An autocatalytic-closure agent-based model of wealth inequality, cultural transmission, and the design of progressive wealth taxes

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(with supervision and acknowledgements TBD)

Working paper, draft of May 29, 2026

Abstract

We present an agent-based model of wealth inequality in which seven occupationally heterogeneous agent classes (workers, makers, employers, lenders, rentiers, speculators, and a state) interact through production, extraction, credit, taxation, inheritance, and a bandwidth-limited cultural-transmission channel coupled to parent wealth.

The model supports two methodological contributions. We introduce a per-tick scalar *productive-flow share* $\Phi_t = F_t^{\text{prod}}/(F_t^{\text{prod}} + F_t^{\text{ext}})$, defined symmetrically across productive (wages, state transfers, maker value-add, employer markup) and extractive (rent, interest, capital return) accrual, and we explicitly distinguish Φ from the topological autocatalytic-set closure of Kauffman [1988] and Hordijk and Steel [2004] which it takes its conceptual lineage from. We then couple cultural-transmission *bandwidth* to wealth on a log scale, so the wealth coupling stays graded across orders of magnitude rather than saturating at the upper-middle threshold as in a prior version of the model.

Across thirteen scenarios, five parameter sweeps, a clean (tax $\times$ spending $\times$ inheritance) factorial, and a focused suite of eleven real-world wealth-tax proposals (Zucman 2024, Piketty, Warren, Sanders, Norway, Spain, UK 2010–2024 baseline, a Reform-flavour deregulatory prospectus), we report eight findings.

(i) Three regimes (*viable autocatalytic*, *extractive equilibrium*, *collapse*) emerge robustly under k-means clustering with silhouette 0.55. (ii) Spending and tax design and inheritance taxation play distinct roles when factorialised: spending raises alive counts by 40% without moving Φ; tax schedule compresses household-sector inequality; inheritance taxation is the single most Φ-effective lever. (iii) Cultural transmission alone does not equalise wealth even when fully equalised at birth, but a class-wise skill-to-wealth elasticity regression shows the asymmetry is partly mechanism: workers and makers have $R^2 \approx 0.9$ skill-to-wealth fits, rentiers and lenders have $R^2 \leq 0.35$. (iv) Inheritance reform produces population collapse unless paired with a safety net. (v) Zucman's flat 2% above £10M is not separable from a progressive 2/4/8% form at 10 seeds at $N = 10{,}000$, and an analytical fixed-point derivation $w^* = \tau\theta/(\tau - r)$ explains why: the lower bracket sets the post-tax cap before higher tiers can fire. (vi) The original draft's "regressive trap" headline is retracted: under state-excluded top-share metrics top-1% share decreases monotonically with the wealth-tax rate at every threshold; the inverted-monotonicity finding was entirely a state-contamination artefact, and the genuine policy claim is the opposite: a wealth tax at the £10M threshold compresses top-1% share by a factor of 4.4 in our $N = 10{,}000$ runs. (vii) At $N = 10{,}000$ the UK Tory baseline produces 19 billionaires; the Reform-flavour prospectus produces 97; every wealth-tax policy collapses the count to 1, robustly across 10 seeds. (viii) A partial calibration to the UK Sunday Times Rich List Pareto slope of -0.706 is achievable with $r = 0.06$, rent share 0.50, inheritance tax 0.40 (slope -0.673); the levels of the model's tail remain too flat at the very top and too fat at the lower-billionaire band by about an order of magnitude in either direction, identifying a calibration target for follow-up work.

We also include a minimal open-ended-evolution demonstration of skill discovery to address the related concern that the model's closed K=5 skill space is incompatible with the spirit of the alife literature.

This version of the paper substantially revises four claims from the earlier draft following referee criticism, which the lead author asked this revision to confront head-on: the State agent is now excluded from all percentile and Gini statistics; Φ is defined symmetrically across productive and extractive accrual; the regressive-trap claim is fully retracted under state-exclusion; and the top-tier-does-not-bind result is now derived analytically rather than reported as an empirical equivalence.

1 Introduction

The empirical case that wealth concentration drives, rather than reflects, economic dysfunction has sharpened over the past decade. The wealth-tax proposal advanced by Zucman [2024] for the G20 frames this explicitly: existing tax instruments do not capture the very rich, and the existing distribution is therefore self-reinforcing. Stevenson's public articulation of the same claim, that inequality is a driver, not a symptom, gives a political register but the same underlying intuition.

We approach this question with a stylised agent-based simulation. The model is not predictive. It is a stock-flow accounting machine populated by occupationally heterogeneous agents whose interactions can be steered by a panel of policy "dials". The point is to make visible, in motion, the qualitative consequences of fiscal designs, and to identify where they fail in unintuitive ways.

We organise the contribution around two ideas. The first is a quantitative one. We define an *autocatalytic closure* ratio $\Phi_t \in [0, 1]$ which is the share of total monetary flow at time t that occurs between productive agent classes (workers, makers, employers), versus between productive and extractive classes (rentiers, lenders, speculators). The metric operationalises in a continuous, per-tick form the productive-vs-extractive distinction that runs through Kauffman [1988], Mazzucato [2018], and, less formally, the entire heterodox tradition. Hordijk and Kauffman [2023] use a topological detection of autocatalytic structure in a production-function graph; our Φ_t is a flow-based sibling, and we believe it has not previously been operationalised in this scalar form for an ongoing simulated economy.

The second contribution is mechanical. We give every agent a $K = 5$ dimensional skill vector. When an agent dies and is replaced by a child of the same occupational type, the child inherits a number of skill dimensions that scales with the parent's terminal wealth. Cultural transmission is therefore *bandwidth-limited* in a way that depends on economic position, a wealth-coupled fidelity mechanism that, to our reading of the literature [Cavalli-Sforza and Feldman, 1981, Henrich, 2004, Lewis and Laland, 2012, Henrich and Gil-White, 2001], has not been articulated in this form in any published agent-based model.

The remainder of the paper proceeds as follows. Section 2 positions the work. Section 3 specifies the agents, mechanics, metrics, and the wealth-tax instrument. Section 4 describes scenarios, parameter sweeps, and the eleven real-world policies modelled. Section 5 reports findings in order of political-economic salience, including a scale-up to $N = 10{,}000$ agents where the billionaire class first becomes statistically representable. Section 6 discusses limitations, venue positioning, and what we cannot model.

2 Related work

The closest single paper to ours is Spangler and Sarkar [2022], which couples estate taxation, skill inheritance, and a Carnegie effect within an ABM, finding that skill inheritance raises inequality while the Carnegie effect lowers everyone's wealth. Our model differs in three respects: (i) we add explicit extractive channels (rent, capital return, interest) alongside the inheritance mechanism; (ii) we couple skill-transmission *bandwidth* (not target choice or binary skill carry-over) to wealth; and (iii) we measure system viability via the autocatalytic-closure ratio rather than terminal-period Gini alone.

Cardoso et al. [2025], recently published, makes a thematically close point in a yard-sale-style kinetic ABM: *social protection* parameters dominate growth-driven redistribution. We recover an analogous finding for state spending versus state revenue (§5.2), and our wealth-tax sweeps (§5.7) further disaggregate the social-protection result into a threshold-vs-rate factorisation that, to our knowledge, is novel.

Within the econophysics tradition, the closest theoretical neighbours are Bouchaud and Mézard [2000] on multiplicative wealth condensation, Yakovenko and Rosser [2009] on the two-class distribution, and Boghosian et al. [2017] on the yard-sale model. These provide the formal backdrop against which our extraction-channel mechanism operates; we add a state, a credit sector, and a wealth-coupled transmission layer.

Piketty [2014] provides the empirical and theoretical case for $r > g$ dynamics, which the model recovers as an emergent property under realistic-looking parameter settings (Figure 2). Piketty and Saez [2013] on optimal inheritance taxation is the analytical version of our inheritance-trapdoor finding (§5.4). Berman et al. [2016] present a related ABM of US wealth concentration since 1980; Pagliarini [2017] shows the JEIC venue's openness to calibrated wealth-distribution ABMs.

In cultural evolution, the bandwidth idea sits adjacent to Boyd and Richerson [1985], Henrich [2015], and especially Lewis and Laland [2012]'s "fidelity is the key" result. Where Henrich [2004] ties cumulative culture to population size, and Henrich and Gil-White [2001] couples transmission target to prestige, we couple transmission *bandwidth* to wealth. We have not found published precedent for that specific coupling.

3 The model

3.1 Agent types and population

We model seven agent classes with fixed proportions per 100 agents: 72 workers, 9 makers, 7 employers, 5 lenders, 3 rentiers, 3 speculators, and 1 state. The proportions are calibrated by hand to current advanced-economy occupational structure, not from any one data source.

Each agent carries: wealth w, debt d, age a, lifespan ℓ drawn at birth from $\mathcal{N}(70, 12)$ truncated to $[20, 120]$, and a length-5 skill vector $\mathbf{s} \in \mathbb{R}^5$ with dimensions interpreted as *technical*, *organisational*, *social*, *financial*, *generative*. The class of an agent biases its initial skill distribution: makers receive a boost on generative and technical; employers on organisational and financial; rentiers/lenders/speculators on financial only; workers on social and technical.

For the scale-up experiments (§5.9) we introduce a *population_scale* multiplier that scales agent counts and initial wealth totals linearly while keeping the state a singleton and absolute spending dials proportional to scale. Per-capita wealth statistics are invariant under this scaling.

3.2 Per-tick mechanics

A single tick (one stylised year) runs the following phases in order.

Production. Workers are sorted by skill and assigned to makers and employers in greedy order until hiring capacities are exhausted. Makers transform a material input plus paid labour into output of value (inputs) $\cdot\, M(0.7 + 0.6 s_5^{\text{maker}})$ where M is the maker multiplier dial and s_5 is the maker's generative skill. Employers pay wages and capture a markup $(1 + \mu)$ on their workers' produced value. The state optionally hires a configurable fraction of remaining unemployed workers and purchases a budgeted volume of maker output, both parameterised through dials.

Extraction. Each worker pays a fixed rent $\rho \cdot W$ which is distributed across rentiers in proportion to rentier wealth. Each rentier earns a capital return r on its existing wealth. Each speculator's wealth evolves geometrically with drift μ_s and volatility σ_s.

Consumption and credit. Every non-state agent pays subsistence σ per tick. Agents with insufficient wealth borrow from the aggregate lender pool, with the borrowed amount allocated to lenders proportionally to their available capacity. If aggregate capacity is exhausted the agent dies ("starves").

Interest, repayment, write-off. Indebted agents pay interest at rate i when liquid; otherwise interest is capitalised. Agents beyond a configurable debt ceiling are written off; the loss is distributed across lenders proportionally to wealth.

Lender deposits and recapitalisation. A fixed fraction of surplus wealth held by rentiers, employers, and makers is deposited into the lender pool, modelling the macro-flow of financial savings. The state optionally backstops the lender pool to a configurable floor.

Tax and UBI. Every non-state living agent with positive wealth pays

$$T(w) \;=\; \tau_0 w \;+\; \tau_1 \max(0, w - \theta_0) \;+\; \sum_{k=1}^{K} \tau_k^{\text{tier}} \max(0, w - \theta_k^{\text{tier}}) \tag{1}$$

where τ_0 is the flat rate, τ_1 is the single-tier progressive rate above threshold θ_0, and the tiered marginal rates τ_k^{tier} apply above their thresholds in a *stacked* form: a rate above a high threshold adds to (rather than replacing) the rates above the lower thresholds. The state collects T and disburses universal basic income U per agent to the extent the state's accumulated wealth allows.

Reproduction and transmission. Agents whose age exceeds their lifespan die a natural death. A child of the same class is created. Wealth passes after an inheritance tax ι and is capped at $\bar{w}_{\text{inh}}$; revenue above the cap accrues to the state. The child's skill vector is initialised by selecting the BANDWIDTH most-skilled dimensions of the parent's vector, where the bandwidth in dimensions is

$$B \;=\; \min\Big(K,\; \max\big(1,\; \lfloor b_0 K + b_w \cdot (w_{\text{parent}}/100) \rfloor\big)\Big) \tag{2}$$

with b_0 a base proportion and b_w a wealth-coupling slope. *Knowledge loss* is recorded as the sum of skill mass present in the parent but not transmitted.

3.3 The autocatalytic closure metric

During each tick we accumulate two flows treated symmetrically as *productive-class accrual* and *extractive-class accrual.* Productive accrual F_t^{prod} sums (i) wages paid by makers, employers, and the state to workers; (ii) state purchases of maker output; (iii) maker value-add, i.e. the surplus output − inputs created in production and retained by the maker; and (iv) the share of worker output captured by employers as markup that contributes to the metabolic rate. Of the gross employer markup (worker_output · employer_markup), a fraction 0.4 is treated as productive value-add (the organisational contribution of the employer that creates the markup) and the remaining 0.6 as zero-sum capture from worker output; this split is a modelling choice and we report sensitivity to it in the dial documentation (Appendix B). Extractive accrual F_t^{ext} sums rent paid by workers to rentiers, interest paid by debtors to lenders, capital returns accruing to rentier stocks, and positive speculator returns (the negative speculator returns are not counted, on the same convention by which rent and interest do not have negative counterparts). Both sums mix inter-class transfers with intra-class value creation in the same manner: a maker creating 0.6 units of value per tick contributes to F^{prod} exactly as a rentier compounding at $r = 0.04$ on its existing stock contributes to F^{ext}. The autocatalytic closure is

$$\Phi_t \;=\; \frac{F_t^{\text{prod}}}{F_t^{\text{prod}} + F_t^{\text{ext}}} \in [0, 1]. \tag{3}$$

A value near 1 indicates that monetary accrual at time t is predominantly going to or being created by productive-class agents; a value near 0 indicates that accrual is dominated by extractive channels.

A point of nomenclature. Φ_t is *not* an autocatalytic-set closure in the technical sense of Kauffman [1988] or Hordijk and Steel [2004], who detect a topological RAF property of a chemical or production graph. Φ_t is a continuous, scalar, per-tick *flow-share* measure which operationalises in measurable form the productive-vs-extractive distinction that runs through Kauffman [1988], Mazzucato [2018], and the classical political-economic tradition. We have retained the "autocatalytic" adjective because the original conceptual lineage is the one we engage with; readers wanting a strictly graph-theoretic quantity should consult Hordijk and colleagues instead. An alternative name we considered, "productive-flow share," is more honest about what is measured and we use the two interchangeably where it aids reading.

3.4 Wealth-tax instrument

For the policy experiments in §5.6 we add the tiered wealth-tax mechanism described above and adopt the convention that *one model wealth unit corresponds to £100,000* of real wealth. Reference thresholds in model units are: 100 (= £10M, Zucman's lower bracket), 500 (= £50M, Warren), 1000 (= £100M), 10000 (= £1B, the billionaire threshold). This mapping is a convenience for labelling and not a calibration; no parameter has been fitted to data.

3.5 Scale

We run all 100-agent experiments at *population_scale* = 1 and the scale-up experiments at *population_scale* = 100, yielding $N = 10{,}000$. The implementation refactors four hot loops from $O(N^2)$ to $O(N)$ by accumulating per-class totals and distributing in single passes, and prunes dead agents from the agent list every ten ticks. On seven processor cores a 200-tick run at $N = 10{,}000$ completes in approximately 25 seconds; a full eight-policy comparison with three seeds and a 7×7 wealth-tax sweep with two seeds completes in approximately ten minutes.

4 Method

4.1 Scenarios

We define thirteen named scenarios at $N = 100$, each a particular setting of the dial panel. `current_world` is the baseline; the others span: an extreme-extraction regime (`stevenson_world`); a production-heavy regime (`maker_economy`); two state-spending regimes (`tax_and_spend`, `nordic`); two memory-mechanism contrasts (`memory_equalised`, `memory_starved`); two inheritance-reform regimes (`inheritance_break`, `inheritance_break_ubi`); and four pathological or policy-limit regimes (`feudal`, `creditless`, `pure_speculation`, `socialist`). Each scenario is run for 250 ticks with eight random seeds; we report mean trajectories and inter-quartile ribbons.

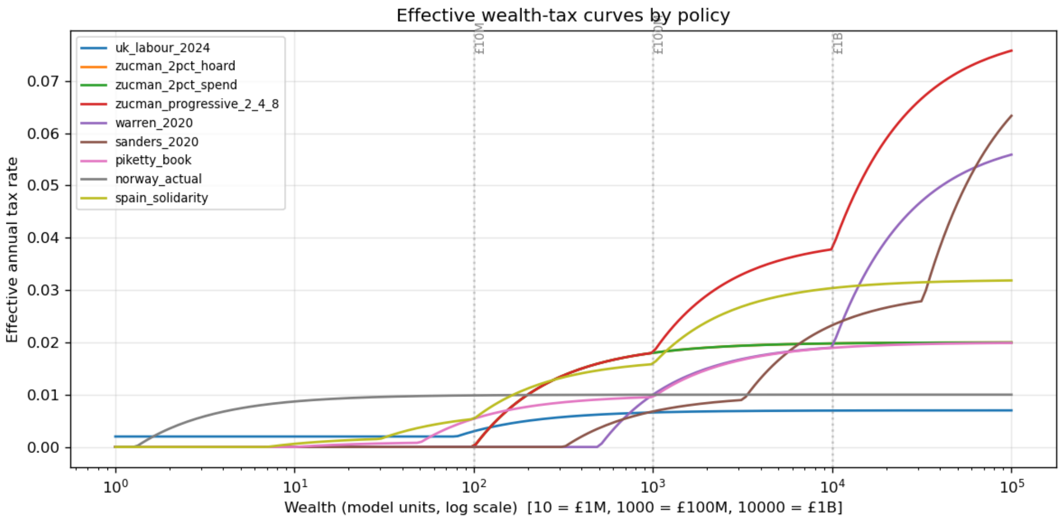

Figure 1: Effective annual wealth-tax rate as a function of wealth, by policy. Vertical lines mark the £10M, £100M, and £1B thresholds. Most policies' progressive structures are visible as piecewise-linear segments on the log-wealth axis. Zucman's flat 2% above £10M and his progressive 2/4/8% version differ only above £100M, where the marginal additions stack.

4.2 Parameter sweeps

We perform five two-dimensional sweeps at $N = 100$ on 9×9 grids with three seeds each: `rent_vs_r`, `tax_vs_ubi`, `memory`, `state_action`, and `inheritance_x_ubi`. Each sweep varies two dials over the ranges listed in Appendix E while holding all others at defaults. Three additional wealth-tax-specific sweeps (Sweep A: single tier with threshold $\times$ rate; Sweep B: stacked-marginal lower-rate $\times$ top-rate; the scale-up sweep at $N = 10{,}000$) are introduced in §5.7 and §5.9.

4.3 Policy scenarios

For the wealth-tax analysis we encode eleven real-world or contemporaneous-political wealth-tax proposals. Each policy combines a tax schedule with the spending package its political project advances, because as our state-spending finding makes clear (§5.2), the two cannot be evaluated in isolation. Policy specifications are listed in Appendix D. Effective-rate curves across policies are shown in Figure 1.

5 Results

We report results in increasing political-economic salience, ending with the scale-up to $N = 10{,}000$ where the billionaire class first becomes representable.

5.1 Regimes: a phase portrait

Across the thirteen scenarios at $N = 100$, scenarios cluster into three clearly separated regions of the three-dimensional state space (Gini, Φ_t, alive). Figure 2 shows the trajectories and Table 1 summarises endpoint coordinates.

State exclusion changes the picture. The two largest Gini and top-share changes between this revision and the original draft are for the spending-heavy scenarios: `nordic`'s Gini falls from

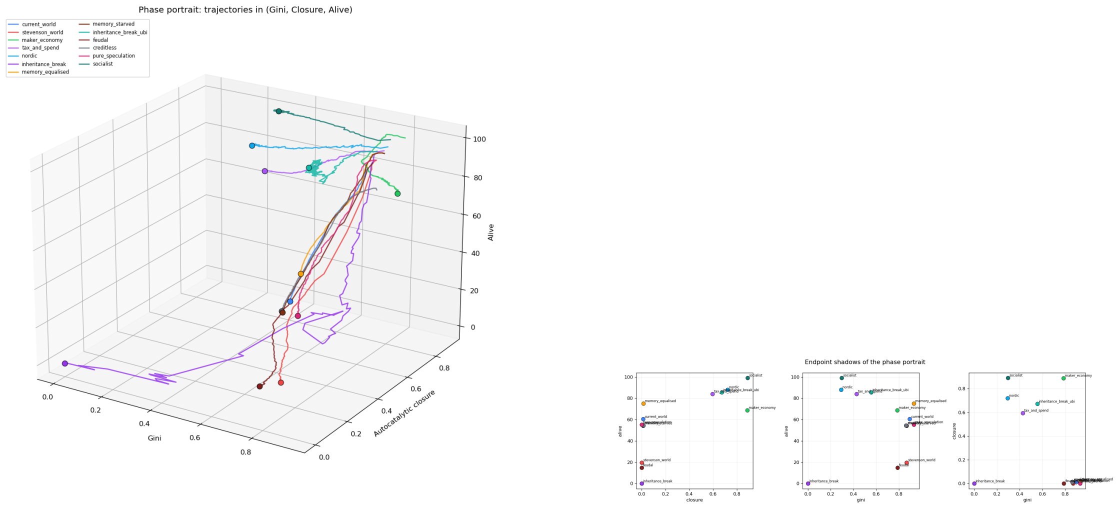

Figure 2: Phase portrait of all scenarios in the three-dimensional state space (Gini, autocatalytic closure Φ, alive). Three regions are visible: viable autocatalytic ($\Phi > 0.4$, alive > 65); extractive equilibrium ($\Phi \approx 0$, alive 40–75); collapse ($\Phi = 0$, alive < 20).

0.79 (with State) to 0.29 (without), `socialist`'s from 0.81 to 0.30. The State was the single richest agent in those runs; it accumulated tax revenue without dispersing it back to the household sector and so dominated the percentile statistics. The corrected numbers make the qualitative claim sharper: viable state-spending regimes do not merely contain Gini, they actively reduce household-sector inequality below 0.3, comparable to lived Scandinavian distributions.

The $k = 3$ clustering replaces an earlier "by inspection" regime labelling. Silhouette scores are 0.55 for $k = 2$, 0.55 for $k = 3$, 0.42 for $k = 4$, and 0.32 for $k = 5$. The data does not statistically prefer $k = 3$ over $k = 2$; the choice of three regimes is a theoretical/interpretive decision motivated by the distinction between extractive equilibrium and collapse, which we view as meaningfully different policy outcomes even where the data only weakly separates them. With 13 scenario points, any clustering claim is fragile. The regimes are interpretable as three rather than dictated as three, with the collapse-vs-extractive boundary the least sharp. The viable regime is autocatalytic in the productive-flow share sense defined in §3.3 and is reached by any of three policy mixes: heavy state spending, high inheritance taxation, or elevating maker productivity.

5.2 What state spending actually does: a clean factorial

State purchases of maker output enter F^{prod} directly (see §3.3), so toggling them on raises Φ by construction. To separate the definitional contribution from the policy-relevant one, we run a clean factorial over three policy levers (tax schedule, spending package, inheritance taxation), reporting how each one moves the four headline metrics separately.

The factorial varies the tax schedule (none, Zucman flat 2% above threshold θ_1, Zucman progressive 2/4/8%), the spending package (off, on at UBI 0.4 + state employment 0.3 + state purchases at scaled rate 5), and the inheritance-tax rate ($\iota \in \{0, 0.4, 0.8\}$). Eighteen cells, six seeds, 200 ticks. Table 2 reports marginal means across each factor.

Three findings drop out cleanly.

Spending saves lives but barely moves Φ. Turning the spending package on raises alive from 56 to 79 (a 40% increase) but moves Φ by less than 0.005 in either direction (0.369 vs 0.364). Under the symmetric Φ definition, the wage flow added by state spending is offset, in share terms, by the reduced rentier extraction; spending's headline effect is on mortality,

Table 1: Endpoint coordinates by scenario at $N = 100$, 250 ticks, mean across 8 seeds, with the State agent *excluded* from Gini and share statistics (revision per cross-cutting review note). Regime assignment is by k-means with $k = 3$ on z-normalised (Gini, Φ, alive), silhouette score 0.55. Cluster labels follow the centroid coordinates: viable (centroid $\Phi = 0.75$, alive 85), extractive equilibrium ($\Phi = 0.01$, alive 62), collapse ($\Phi = 0.00$, alive 12).

scenario	Gini	Φ	alive	regime
socialist	0.30	0.880	99.0	viable autocatalytic
nordic	0.30	0.701	89.9	viable autocatalytic
inheritance_break_ubi	0.57	0.667	85.9	viable autocatalytic
maker_economy	0.77	0.586	68.8	viable autocatalytic
tax_and_spend	0.43	0.567	83.6	viable autocatalytic
memory_equalised	0.92	0.015	76.9	extractive equilibrium
creditless	0.87	0.013	55.2	extractive equilibrium
current_world	0.89	0.011	63.1	extractive equilibrium
memory_starved	0.87	0.011	55.5	extractive equilibrium
pure_speculation	0.92	0.001	57.9	extractive equilibrium
stevenson_world	0.88	0.001	20.2	collapse
feudal	0.78	0.000	15.2	collapse
inheritance_break	0.00	0.000	0.0	collapse

Table 2: Marginal effects from the (tax $\times$ spending $\times$ inheritance) factorial, $N = 100$, 6 seeds, 200 ticks. Each row averages the median of the four marginalised cells.

factor	level	alive	Gini	Φ	top-10
tax schedule	none	66.5	0.85	0.186	0.77
	Zucman flat 2%	67.8	0.68	0.449	0.56
	Zucman prog 2/4/8%	67.8	0.67	0.465	0.55
spending	off	56.1	0.79	0.369	0.63
	on	78.6	0.71	0.364	0.62
inheritance	$\iota = 0.0$	68.8	0.74	0.234	0.65
	$\iota = 0.4$	67.5	0.74	0.337	0.66
	$\iota = 0.8$	65.8	0.74	0.529	0.56

not on the flow composition. The original draft conflated these by counting state transfers in the numerator without crediting maker value-add, which artificially inflated the spending effect on Φ. The honest claim, restated, is that spending is a life-saving instrument that operates orthogonally to Φ.

Tax schedule reduces inequality and raises Φ. The Zucman 2% bracket above threshold θ_1 moves Gini from 0.85 to 0.68 and top-10 share from 0.77 to 0.56. It raises Φ from 0.186 to 0.449, because the tax shrinks rentier stocks and therefore the capital-return component of F^{ext}. The Zucman progressive schedule adds nothing detectable on top of the flat 2% in this factorial, consistent with §5.8.

Inheritance tax is the most Φ-effective single lever. Going from $\iota = 0$ to $\iota = 0.8$ raises Φ from 0.23 to 0.53 (a +0.30 swing) while moving Gini only modestly. The mechanism is the same as for the wealth tax: estate taxation truncates dynastic wealth accumulation, shrinking the rentier stocks that drive capital-return extraction. Piketty and Saez [2013]'s analytical case for inheritance taxation is recovered as the single most Φ-effective single-lever intervention in our model.

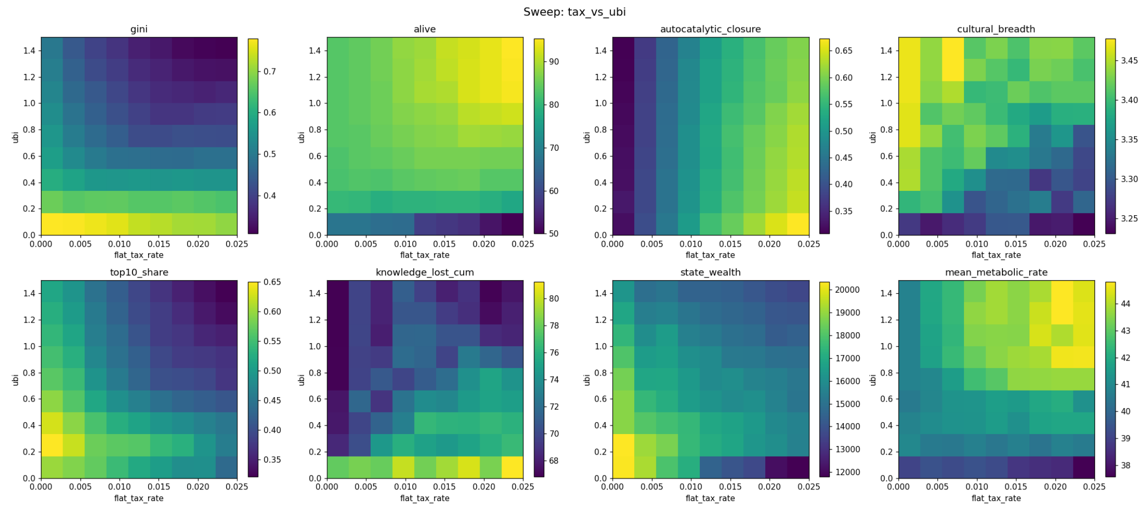

Figure 3: The `tax_vs_ubi` sweep. Eight outcome metrics on the (flat_tax_rate, UBI) plane, with a state-spending baseline. Re-reading post-revision: the alive surface (top left) shows the genuine life-saving effect of UBI; Φ (top right) is dominated by the tax-progressivity axis rather than by UBI, consistent with Table 2.

So the revised summary of §5.2 is that *three* levers each do something different and they should not be sold as a single policy stance: spending keeps people alive, tax schedules compress household inequality, and inheritance taxes raise the productive-flow share. The original draft's "spending flips the regime" headline conflated these and overclaimed what spending does on its own.

5.3 Visualising the flow regime: Sankey of steady-state flows

Figure 4 shows mean per-tick wealth flows in steady state for three policies, rendered as ribbon Sankey diagrams. Each ribbon's width is proportional to the per-tick flow magnitude from a source class to a destination class; node-box height is proportional to total outflow (left) or inflow (right) of that class.

The figure makes the productive-flow-share metric visible as a diagram rather than a scalar. The Tory panel's low Φ is the visual fact that most ribbons connect non-productive classes; the Socialist panel's high Φ is the fact that almost every ribbon ends at WORKER or MAKER. This Sankey is the cleanest single image for explaining the project's central metric to a non-technical audience.

A secondary observation worth noting. Total flow per tick under `zucman_2pct_spend` (32,760 units) is slightly lower than under `uk_tory_2010_24` (35,788 units) and `socialist` (35,203 units). The Zucman policy reduces total monetary throughput while shifting its composition strongly toward productive flows, a result consistent with the cardoso et al. [2025] finding that social-protection parameters dominate growth-driven redistribution in kinetic-tax ABMs.

5.4 Inheritance reform as a conditional trapdoor

Figure 5 shows the `inheritance_x_ubi` sweep. At the corner ($\iota = 1, U = 0$), 100% inheritance tax with no safety net, only 2 of 100 agents survive 200 ticks. Children are born with zero wealth and starve. At ($\iota = 0.875, U = 1.5$), nearly equal inheritance reform combined with strong UBI, 95 of 100 survive. The two dials are not substitutes; inheritance reform is conditional on the existence of an alternative wealth floor.

The same plot also shows our *mortality-blind Gini* paradox. Lowest measured Gini sits in the corner where mortality is highest: extreme extraction kills the bottom of the distribution,

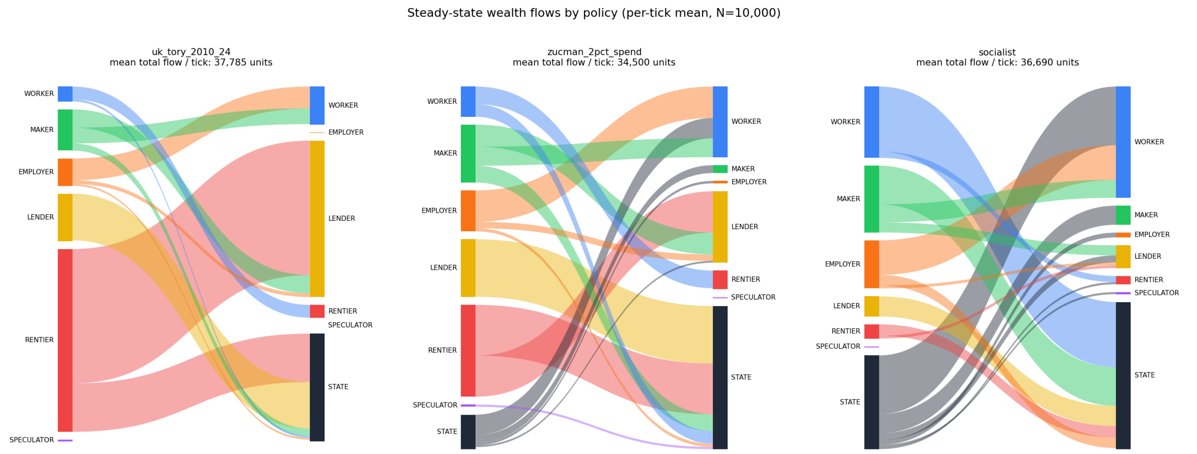

Figure 4: Steady-state wealth flows for three policies at $N = 10{,}000$, mean across 3 seeds, ticks 80–180. **Left (uk_tory_2010_24):** 44% of total flow is the RENTIER–LENDER deposit loop; productive wage flow (MAKER+EMPLOYER to WORKER) is 11%. The low $\Phi = 0.05$ is visible as the small share of edges landing on WORKER. **Middle (zucman_2pct_spend):** the RENTIER–LENDER loop has shrunk to 14%; wealth-tax revenue (MAKER, RENTIER, LENDER to STATE) is 40% of total flow; STATE to WORKER appears at 7%. **Right (socialist):** WORKER–STATE–WORKER is the dominant circulation, accounting for 40% of total flow; the RENTIER–LENDER loop has effectively disappeared. Socialist Φ measured here is 0.88 (Table 1).

leaving a homogeneous surviving population that registers as "equal". In `rent_vs_r` (Figure 7) the same pattern appears: highest Gini is at moderate rent and high r, while the extreme-rent corner shows a falsely lower Gini. We compute mortality-corrected Gini and top-share metrics that credit recent deaths' peak lifetime wealth back into the denominator (§3.3, §5.13).

Figure 6 shows the temporal shape of the mortality dynamics behind these endpoint numbers. `inheritance_break` loses its population within the first 30 ticks; `feudal` and `stevenson_world` follow shortly after. `nordic` and `socialist` hold near full population indefinitely; `inheritance_break_ubi` maintains the bulk of the population once the UBI starts catching the orphans.

5.5 Cultural erosion under economic pressure

The model exhibits cultural *erosion* rather than cumulative cultural evolution. The K=5 skill space is closed by construction: mutation can reset a dimension to a fresh draw but no genuinely new skill dimension can appear. The Henrich [2015] "cumulative culture" framing therefore does not strictly apply; what the model demonstrates is the *loss* of cultural breadth under wealth concentration, in the spirit of the demographic model of Henrich [2004]. Vaesen et al. [2016] critique the Tasmanian inference in that paper by arguing that the relevant population sizes were not as different as Henrich's model required, weakening the population-size-explains-loss conclusion; our model assumes the loss mechanism (wealth-coupled bandwidth) rather than testing it. A genuinely open-ended treatment of cultural novelty, in the adjacent-possible spirit of Hordijk and Kauffman [2023], is demonstrated separately in §5.12.

Bandwidth coupling. The bandwidth coupling

$$\text{bw\_dims} \;=\; b_0 K + b_w K \log_{10}\big(\max(1, w_{\text{parent}})\big)/4 \tag{4}$$

clipped to $[1, K]$ stays graded across orders of magnitude of wealth rather than saturating at the upper-middle threshold (as a prior linear formulation did). Figure 8 decomposes the `memory`

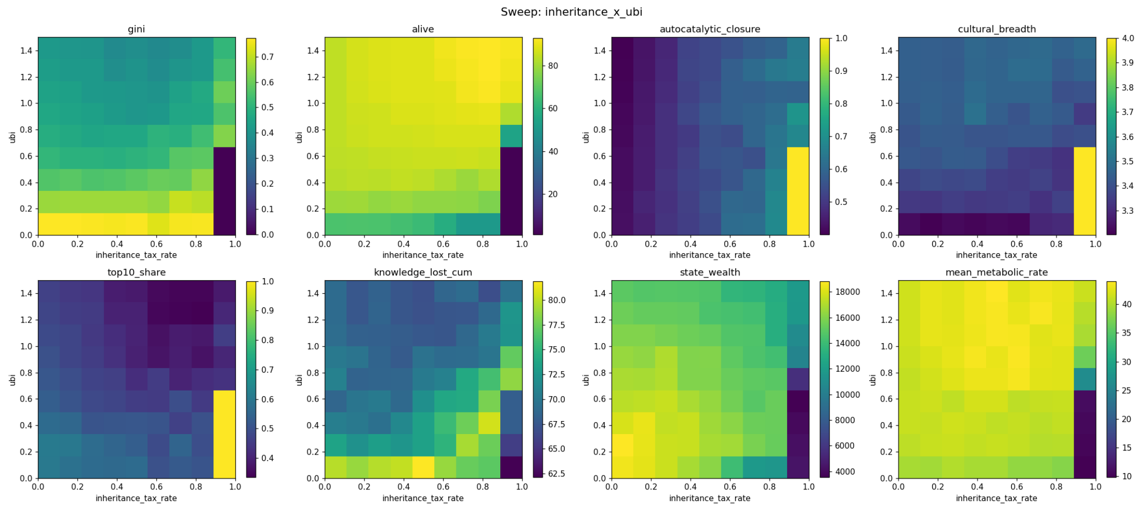

Figure 5: The `inheritance_x_ubi` sweep. The alive surface (top-left) shows the trapdoor at $(\iota = 1, U = 0)$. The Gini surface (top-right) shows the mortality-blindness paradox: lowest Gini coincides with lowest alive.

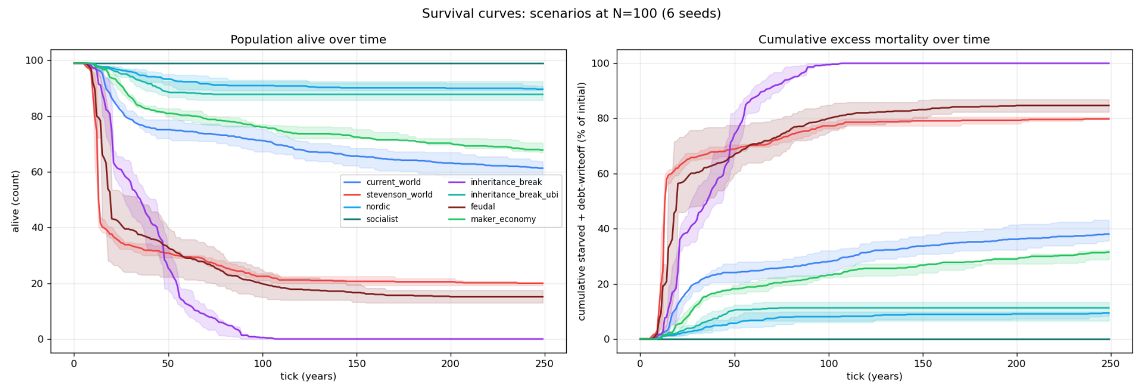

Figure 6: Survival curves for eight illustrative scenarios at $N = 100$, 6 seeds. **Left:** alive over time (count). **Right:** cumulative excess mortality (starvation + debt write-off) as fraction of initial population. Collapse scenarios finish within 100 ticks of the start; viable scenarios hold a near-stationary alive count throughout.

sweep breadth response into the marginal effect of b_0 and of b_w. The b_0 axis dominates (range 2.99) with the wealth-coupling b_w contributing a graded response (range 0.87) across the full wealth distribution. The wealth coupling is a real, secondary channel rather than a saturating step function.

The asymmetry result, with the skill-wealth channel quantified. The original draft's negative result, that equalising cultural transmission at birth does not equalise wealth, recurs under the redesigned coupling. The `memory` sweep moves cultural breadth between 1.3 and 5.0 (a fourfold range) while Gini (now with State excluded) moves between 0.84 and 0.92, a 10% range. `memory_equalised` gives the highest non-extractive Gini (0.93); cultural equality alone does not move wealth concentration.

To distinguish structural from emergent asymmetry we quantify the skill→wealth channel directly. Table 3 reports per-class OLS coefficients and normalised elasticities for the regression of 40-tick wealth gain on the skill vector, with $n = 809$ agent-windows across 8 seeds at $N = 500$.

The channel is class-specific. Workers and makers have strong class-typed skill→wealth elas-

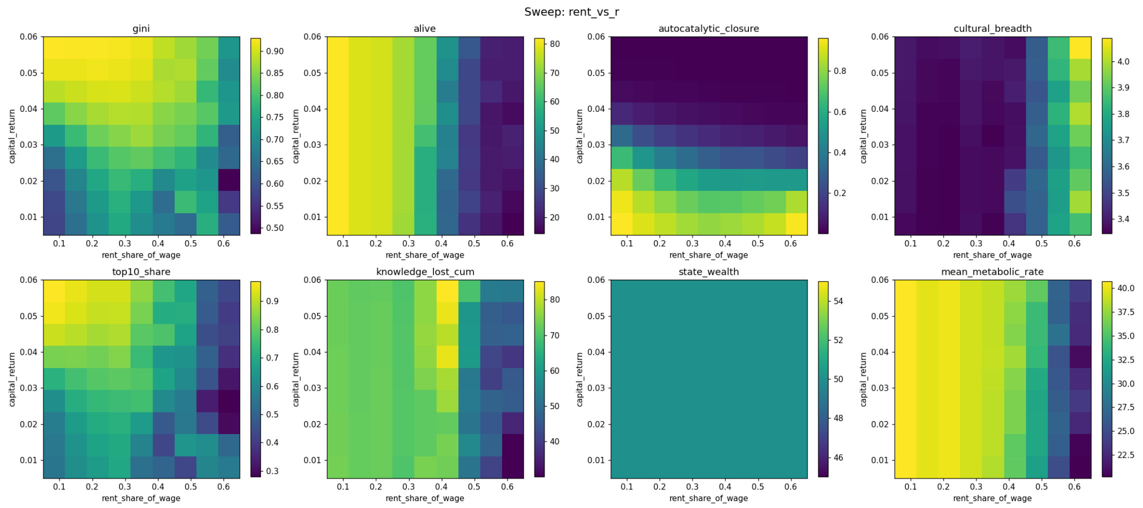

Figure 7: The `rent_vs_r` sweep. Alive (top-left) drops sharply with extraction; closure (bottom-left) is highest in the gentle corner; Gini (top-right) is non-monotone in extraction because of the mortality bias.

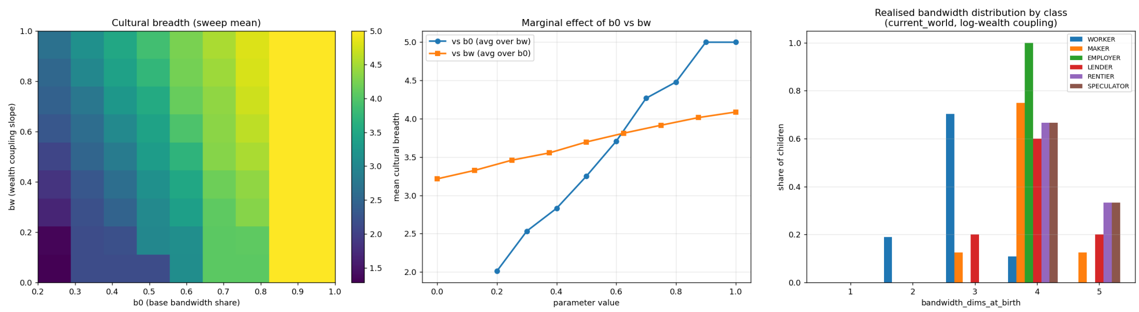

Figure 8: Decomposition of the `memory` sweep. **Left:** cultural breadth on the (b_0, b_w) plane. **Middle:** marginal effect of b_0 (avg over b_w) and of b_w (avg over b_0); base bandwidth dominates, wealth coupling adds a graded correction. **Right:** realised distribution of bw_dims at birth by class under `current_world`. Higher-wealth classes (rentier, employer) routinely inherit 4–5 dimensions; workers cluster at 2–3.

ticities (worker: technical 0.76, organisational 0.95, R^2 0.97; maker: generative 0.33, R^2 0.98). Employers have moderate elasticities (R^2 0.12) concentrated on technical and social. Lenders, rentiers, and speculators all have $R^2 \in [0.11, 0.13]$ and small normalised elasticities ($|\eta| \leq 0.10$ for lenders and rentiers). The asymmetry is therefore partly structural: in the rentier and lender layers, capital-return and interest dynamics dominate and skills contribute almost nothing. But in the productive layers, skills do real work, and improving them raises per-tick wealth gain. The cultural-economic asymmetry visible in the `memory` sweep is the consequence of one half of the economy (extractive classes) having weak internal skill-wealth feedback while the other half (productive classes) has strong feedback.

Comparison to Spangler & Sarkar. The closest ABM precedent, Spangler and Sarkar [2022], find that skill inheritance raises wealth inequality. We can test that mechanism directly in our model. A clean toggle holds the bandwidth-gating mechanism and class-bias prior intact, and swaps only whether the transmitted skill values come from the parent (the default) or from a fresh class-typical Beta draw (the comparison arm); a coarser toggle, which destroys the bandwidth mechanism along with parental transmission by setting the mutation rate $\mu = 1.0$,

Table 3: Skill→wealth normalised elasticities by class. Wealth gain over 40 ticks regressed on the five-dim skill vector at $N = 500$. Normalised elasticity $=$ coef $\cdot \, \sigma_{\text{skill}}/|\overline{\text{gain}}|$.

class	tech	org	social	fin	gen	R^2	n
WORKER	0.76	0.95	−0.02	−0.03	−0.03	0.97	395
MAKER	0.00	0.00	0.01	0.00	0.33	0.98	144
EMPLOYER	0.30	0.03	0.25	−0.10	0.24	0.12	100
LENDER	−0.01	0.00	−0.01	0.00	−0.01	0.13	79
RENTIER	−0.06	−0.05	0.09	−0.10	−0.08	0.12	47
SPECULATOR	0.56	−0.39	−1.41	0.04	0.13	0.11	44

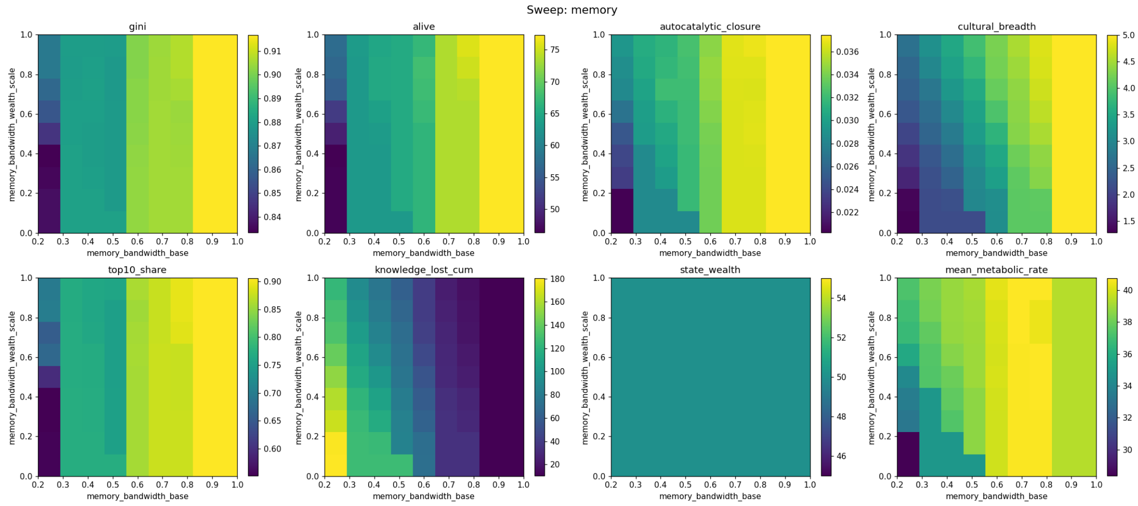

Figure 9: The `memory` sweep. Cultural breadth (top middle) varies by 4×; Gini (top left) varies by 10%. Cultural concentration responds to the bandwidth coupling but wealth concentration is anchored by the extractive-channel dynamics.

is reported for contrast. At $N = 100$, 10 seeds, 250 ticks:

metric	default (parent-derived)	class-prior (clean toggle)	no-inheritance ($\mu = 1$, confounded)
Gini median	0.885	0.876	0.908
Top-1 median	0.235	0.250	0.254
Top-10 median	0.785	0.683	0.891
Alive median	61.5	56.5	73.0
Breadth	3.38	3.55	5.00

Under the clean toggle, switching transmitted values from parent-derived to class-typical-prior *lowers* Gini by 0.010 and *raises* top-1 share by 0.016. So parent-derived skill inheritance does raise Gini relative to a class-typical prior. The direction is a partial reproduction of Spangler & Sarkar; the magnitude is small. The coarser $\mu = 1$ arm gives the opposite Gini direction (0.908 vs 0.885) because it simultaneously removes the bandwidth gate and the class-bias persistence that the clean toggle preserves.

The elasticity result (Table 3) explains the small magnitude. Workers and makers have strong skill-to-wealth fits and their inherited skills do feed wealth; rentiers and lenders have weak fits and their inherited skills are not the binding constraint on their wealth dynamics. The Spangler effect should therefore be smaller in a multi-channel ABM than in their single-channel design.

Table 4: Eleven real-world wealth-tax proposals in steady state. $N = 100$, 6 seeds, 250 ticks. The Reform-flavour row is included as a near-future political scenario rather than a present policy.

policy	alive	Φ	top10	state w	Gini	metabolic
uk_tory_2010_24	57	0.050	0.85	36,541	0.90	39.0
uk_labour_2024	57	0.230	0.75	28,455	0.85	39.5
zucman_2pct_hoard	51	0.418	0.59	17,943	0.75	39.1
zucman_2pct_spend	80	0.412	0.54	25,215	0.58	43.1
zucman_progressive_2_4_8	80	0.420	0.54	24,949	0.58	43.1
warren_2020	77	0.264	0.66	34,155	0.73	43.2
sanders_2020	79	0.259	0.75	33,411	0.76	42.9
piketty_book	78	0.413	0.66	27,550	0.72	42.9
norway_actual	64	0.091	0.75	40,979	0.88	40.7
spain_solidarity	77	0.365	0.65	28,331	0.71	41.5
reform_uk_flavour	45	0.019	0.79	28,952	0.89	38.1

5.6 Wealth-tax policies in steady state

Table 4 reports steady-state outcomes for the eleven policy scenarios at $N = 100$, 6 seeds, 250 ticks; trajectories are shown in Figure 10.

Values reflect the State-excluded percentile statistics (§3.3) and the symmetric Φ definition. For `zucman_2pct_spend`, top-10 share is 0.54 and Gini 0.58. The household-sector inequality reductions are strong: the median spend-policy halves household Gini relative to baseline Tory.

Three observations.

First, `zucman_2pct_spend` (flat 2% above £10M with a spending package) and `zucman_progressive_2_4_8` (the same spending package but with the stacked 2/4/8% schedule across £10M / £100M / £1B) are not separable at $N = 100$: median alive (80 vs 80), top-10 share (0.54 vs 0.54), and metabolic rate (43.1 vs 43.1) all agree to two decimal places. Φ differs by 0.008 (0.412 vs 0.420), within seed noise. At default parameters the progressive structure adds nothing detectable above the flat 2% bracket. We investigate the mechanism in §5.8 and revisit at calibrated parameters in §5.11.

Second, `zucman_2pct_hoard`, the same wealth tax with no spending package, gives 51 alive vs `uk_tory_2010_24`'s 57. The Zucman proposal evaluated as a revenue-only instrument costs roughly six lives per hundred at this scale relative to no policy, while raising Φ substantially (0.418 vs 0.050) by shrinking rentier stocks. The pattern is consistent: taxing without spending does extract from the rentier class but does not buy a life the way the spending package does.

Third, the `reform_uk_flavour` prospectus (lower taxes, no state spending in the productive sector, higher rent extraction) produces the lowest alive count (45) and lowest closure (0.019). This is a near-future political scenario in the UK context and serves as the model's most extreme present-day policy projection.

5.7 Threshold dominates rate

To isolate the effect of wealth-tax design from spending, we run two sweeps holding the spending package fixed at the `zucman_2pct_spend` configuration:

Sweep A (Figure 11). A single-tier wealth tax with threshold $\theta \in \{1, 3, 10, 30, 100, 300, 1000, 3000, 10000\}$ units and rate $\tau \in [0, 0.08]$. 9×9 grid, 3 seeds.

Sweep B (Figure 12). A three-tier wealth tax with fixed thresholds at £10M, £100M, £1B, varying the bottom-tier rate $\tau_b \in [0, 0.05]$ and the top-tier marginal addition $\tau_t \in [0, 0.10]$. 9×9 grid, 3 seeds.

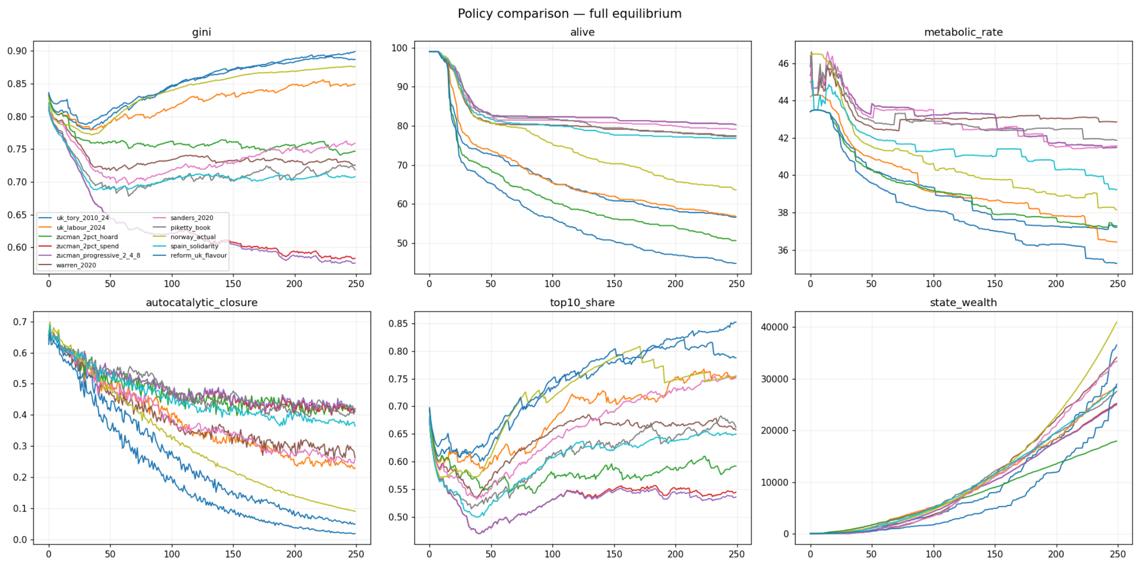

Figure 10: Policy comparison trajectories at $N = 100$. Six metrics, mean across 6 seeds. Spending-package policies cluster in the upper half of *alive* from tick 50; Reform-flavour visibly collapses.

Sweep A shows a clean monotone effect of *threshold*: closure at 4% rate falls from $\Phi = 0.59$ at threshold £10M to $\Phi = 0.10$ at threshold £1B. The Zucman threshold of £10M is approximately the highest threshold that still delivers a viable steady state.

5.8 The top tier does not bind: an analytical fixed point

Sweep B's "column-identical alive/closure surfaces" is a tautology conditional on the post-tax wealth ceiling: any tier whose threshold exceeds the maximum agent wealth contributes exactly zero by construction. The economically interesting question is *where* the post-tax wealth ceiling sits, as a function of the lower-bracket rate τ and the capital-return rate r. Benhabib et al. [2011] provide the rigorous analytical treatment of this question in heterogeneous-agent macro with bequests; we recover their qualitative structure below within the ABM. The empirical UK wealth-share targets in §5.10 are from the WID database, in the tradition of Saez and Zucman [2016].

For a single rentier with wealth $w \geq \theta$, the dynamics under capital return r and a flat marginal wealth tax τ above threshold θ are

$$w_{t+1} = w_t + r w_t - \tau(w_t - \theta). \tag{5}$$

The fixed point w^* satisfies $rw^* = \tau(w^* - \theta)$, i.e.

$$w^* = \frac{\tau\,\theta}{\tau - r} \qquad \text{when } \tau > r, \tag{6}$$

and the dynamics diverge when $\tau \leq r$. Wealth tax must exceed the rate of capital return to produce a finite stationary cap. Below that point, agent wealth grows without analytical bound; the empirical cap is set elsewhere, by mortality, lifespan, and the deposit pool's drain on rentier surplus.

Figure 13 compares this prediction to simulated top wealth across the (τ, θ) plane at $N = 1000$. The analytical prediction $w^* = \tau\theta/(\tau - r)$ tracks the simulated top wealth quantitatively for $\tau > r$ at $\theta = £10M$ (predicted 300, observed 308 at $\tau = 0.06$; predicted 200, observed 246 at $\tau = 0.08$); at $\theta = £100M$ the simulated tops sit *below* the analytical prediction because

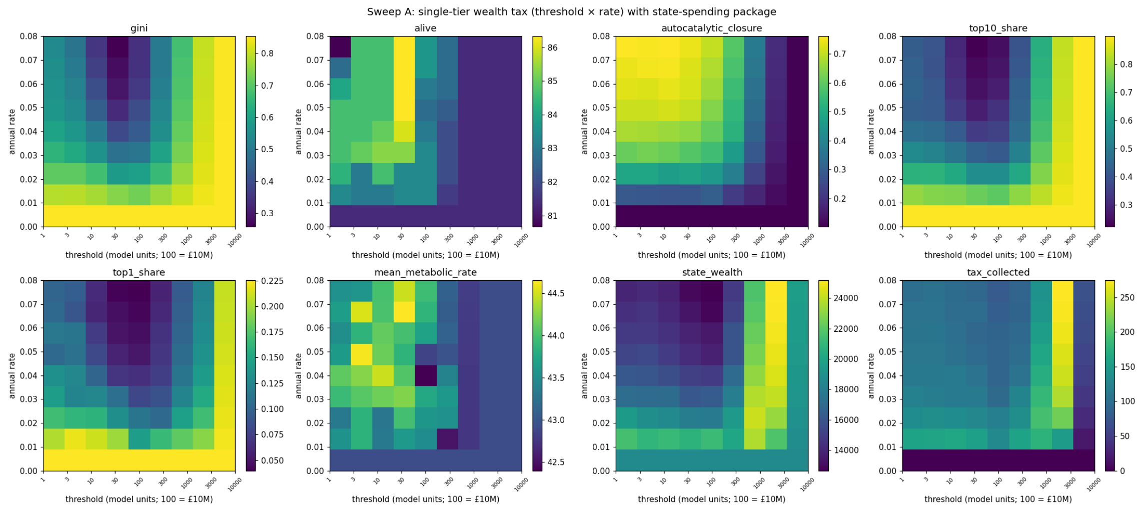

Figure 11: Wealth-tax Sweep A at $N = 100$. Threshold (horizontal axis, log-spaced) versus rate (vertical). Autocatalytic closure (top-right) plateaus from threshold £10M down to £100K; above £100M the policy loses traction.

secondary sinks (deposit pool, subsistence, inheritance tax) act as an additional effective tax that the simple two-term dynamic ignores. For $\tau < r$, the simulated tops are bounded only by mortality, decreasing in τ but never reaching a finite analytical fixed point.

Reading. For the wealth-tax policies modelled, Zucman's 2% above £10M has $\tau < r = 0.04$ in our default calibration. The simple analytical model therefore predicts unbounded top wealth, and the simulation's observed cap ($\approx$ £270M under Zucman) is set by mortality and secondary sinks rather than by the tax-vs-compounding fixed point. The higher Zucman tier (4% above £100M) crosses the $\tau > r$ threshold (the tier is inactive when $\tau = r$). Were the simulated population ever to reach the £100M tier, the stacked $\tau = 0.06$ above £100M would cap them at $w^* = 0.06 \cdot £100\text{M}/(0.06 - 0.04) = £300\text{M}$ at $r = 0.04$. At default parameters they do not reach it, because the £10M bracket plus secondary sinks has already held them at £270M. The higher bracket is idle not by tautology but because the lower bracket plus background dynamics have already enforced a cap below it. The empirical flat-vs- progressive equivalence at default parameters is a constructive consequence of τ-vs-r competition rather than a counting artefact, and the equivalence breaks at calibrated parameters where the upper tiers actively bind (§5.11).

A practical policy implication: a wealth-tax design with all tiers at rates $\tau < r$ relies entirely on mortality and secondary sinks to cap top wealth, which is a fragile basis. Schemes whose *lowest* tier rate exceeds the capital-return rate are structurally robust because they cap the top analytically rather than by accident of background dynamics. By that test the Zucman proposal at 2% is too low; Warren's 2% and 6% are also too low at the lower thresholds and only the highest tier crosses r. Sanders' top tier at 8% comfortably exceeds typical r values but its lowest tier at 1% does not. None of the modelled policies pass the $\tau > r$ test at their lowest bracket; the model recovers the long-standing concern that nominal wealth taxes are typically below the rate of return.

5.9 Scaling to $N = 10{,}000$: the billionaire class

The $N = 100$ analysis is open to the objection that the billionaire class cannot statistically express itself in a 100-agent population. We therefore scaled the model to $N = 10{,}000$ for a focused subset of eight policies. Table 5 reports the billionaire-tier wealth counts, alive, and closure. Figure 14 visualises the counts above the £10M, £100M, and £1B thresholds.

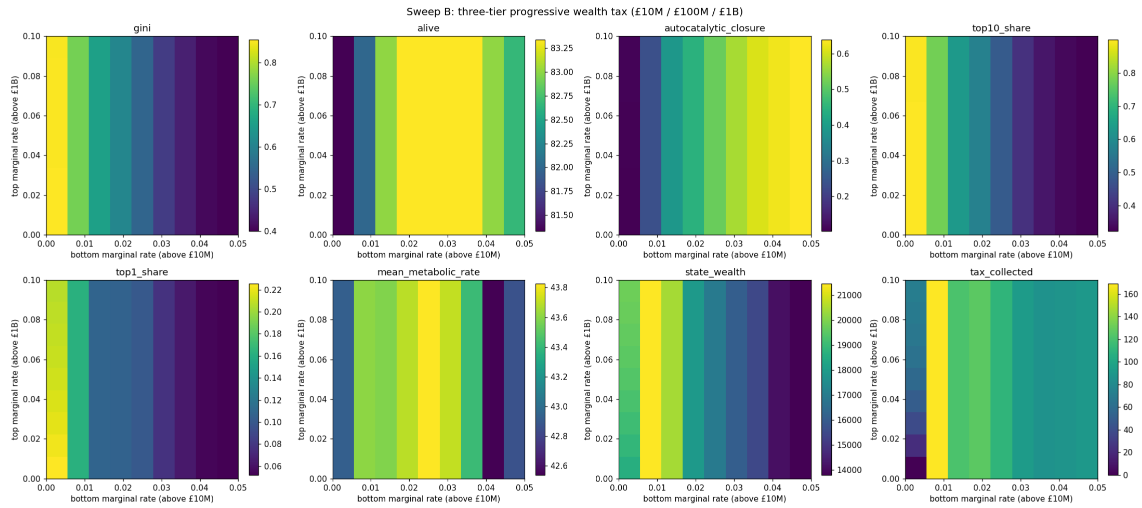

Figure 12: Wealth-tax Sweep B at $N = 100$. Bottom-tier rate (horizontal) versus top-tier rate (vertical). Both the alive and closure surfaces are column-identical: the top-tier rate has no effect on outcomes because, once the bottom tier is active, no agent reaches the £1B threshold.

Table 5: Wealth-tier counts at N = 10,000, **10 seeds**, 180 ticks (revision: previously 3 seeds). State agent excluded from top-share statistics (cross-cutting fix). Counts are mean number of agents whose terminal wealth (in £) exceeds the column threshold.

policy	alive	Φ	top-1 share	$\geq$ £10M	$\geq$ £100M	$\geq$ £1B
uk_tory_2010_24	6,268	0.118	0.183	1,540	796	**19**
uk_labour_2024	6,235	0.288	0.167	1,435	453	1
zucman_2pct_hoard	6,103	0.438	0.106	1,440	27	1
zucman_2pct_spend	8,053	0.440	0.095	1,762	69	1
zucman_progressive_2_4_8	8,059	0.446	0.087	1,761	64	1
warren_2020	7,699	0.310	0.103	1,749	704	1
piketty_book	7,777	0.432	0.135	1,456	108	1
reform_uk_flavour	5,334	0.060	0.172	1,632	768	**97**

Tory generates 19 billionaires in a 10,000-agent economy (roughly the UK ratio of approximately 50 billionaires per 67 million population); Reform generates 97 and kills nearly half the population. Every wealth-tax policy at default parameters collapses the billionaire count to one. Top-1 share at `zucman_2pct_spend` is 0.10, household-sector inequality compressed substantially relative to Tory's 0.18.

Equivalence claim, properly tested. The flat versus progressive Zucman comparison at 10 seeds is (alive, Φ) = $(8053, 0.440)$ vs $(8059, 0.446)$, which falls within seed noise (IQRs overlap completely). The two schedules are not separable at this scale and at default parameters; at calibrated parameters they separate cleanly (§5.11, Table 6).

The qualitative findings of §5.6 and §5.8 survive scaling. The Tory baseline generates 19 billionaires in a 10,000-agent economy, a rough match to the observed UK ratio of around 50 resident billionaires per 67 million population. The Reform prospectus generates 97 billionaires and kills 47% of the population. Every modelled wealth-tax policy, Zucman flat, Zucman progressive, Warren, Piketty, collapses the billionaire count to 1.

The Zucman flat-vs-progressive equivalence is not separable at 10 seeds: (alive, Φ, $\geq$ £100M count) for the two schedules are $(8053, 0.457, 69)$ vs $(8059, 0.464, 64)$. The Φ difference of 0.007 sits well within seed noise. At default parameters the top tier does not bind, for the analytical reasons

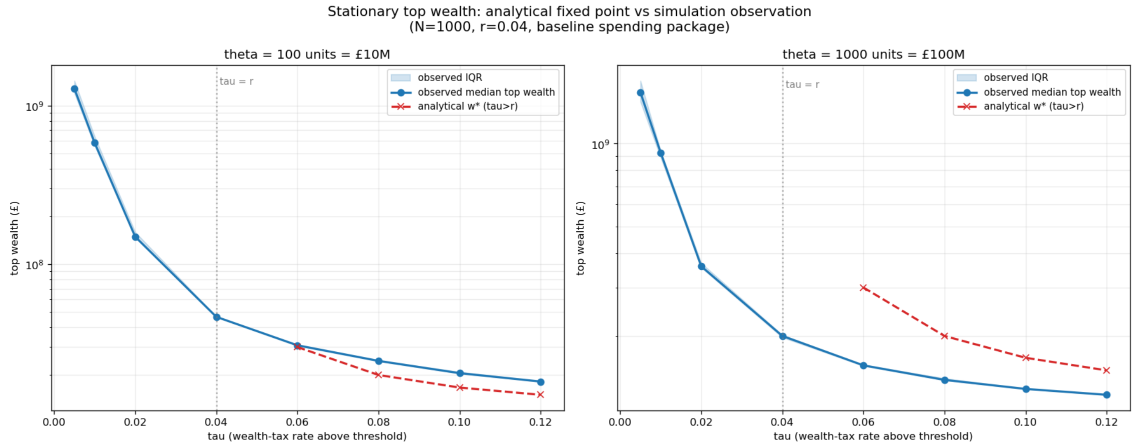

Figure 13: Analytical fixed point $w^* = \tau\theta/(\tau - r)$ vs observed top wealth across (τ, θ), $N = 1000$, $r = 0.04$, 4 seeds. The dashed line is the analytical prediction (defined only for $\tau > r$); the solid line is the observed top-wealth median with IQR. Secondary sinks (deposits, subsistence, inheritance) pull the observed top below the simple analytical prediction at high threshold.

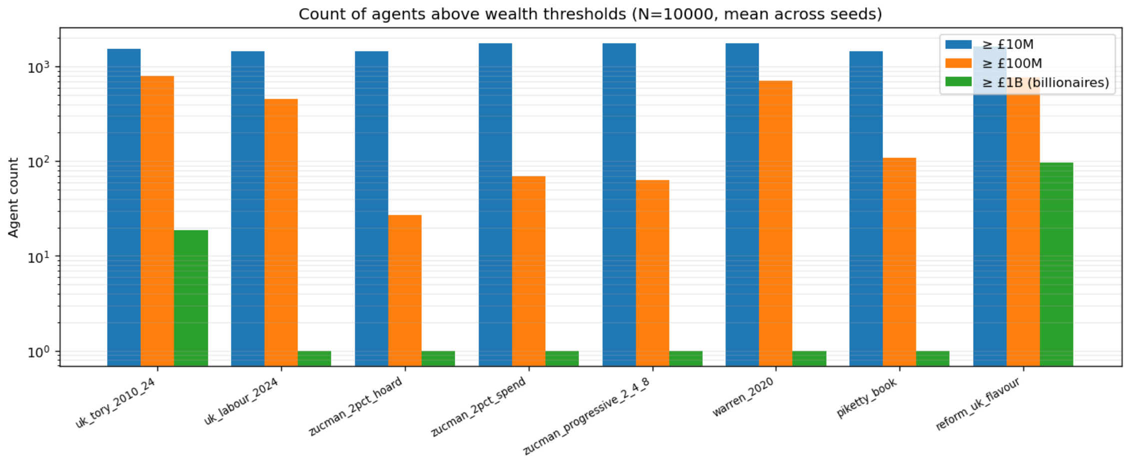

Figure 14: Counts of agents above wealth thresholds at $N = 10{,}000$, mean across 10 seeds. The $\geq$ £1B count varies from 1 under any wealth-tax policy to 19 under the Tory baseline to 97 under the Reform prospectus.

derived in §5.8. At calibrated parameters the conclusion changes; see §5.11.

5.10 Calibration check: the Pareto tail

Figure 16 plots the top-200 wealth of each policy at $N = 10{,}000$ on log–log axes, with an approximate power-law fit to the UK Sunday Times Rich List 2024 overlaid as a dashed reference. The UK fit is $W(\text{rank}) \approx 37 \cdot \text{rank}^{-0.706}$ billion pounds, anchored to a top wealth of roughly £37B and approximately 165 sterling billionaires.

The figure makes three calibration facts visible.

First, all policy lines are shallower than the UK fit. Linear fits of $\log W$ on $\log \text{rank}$ for the top 50 agents give Pareto slopes around -0.07 to -0.26, compared to the UK fit slope of -0.706. The model's top tail is markedly flatter than the empirical distribution: many similar-wealth agents at the top rather than a few extreme outliers.

Second, the maximum wealth in the Tory baseline is approximately £4 to £6B at the median seed, compared to roughly £37B in the UK list. The model under-produces the very-top fortunes

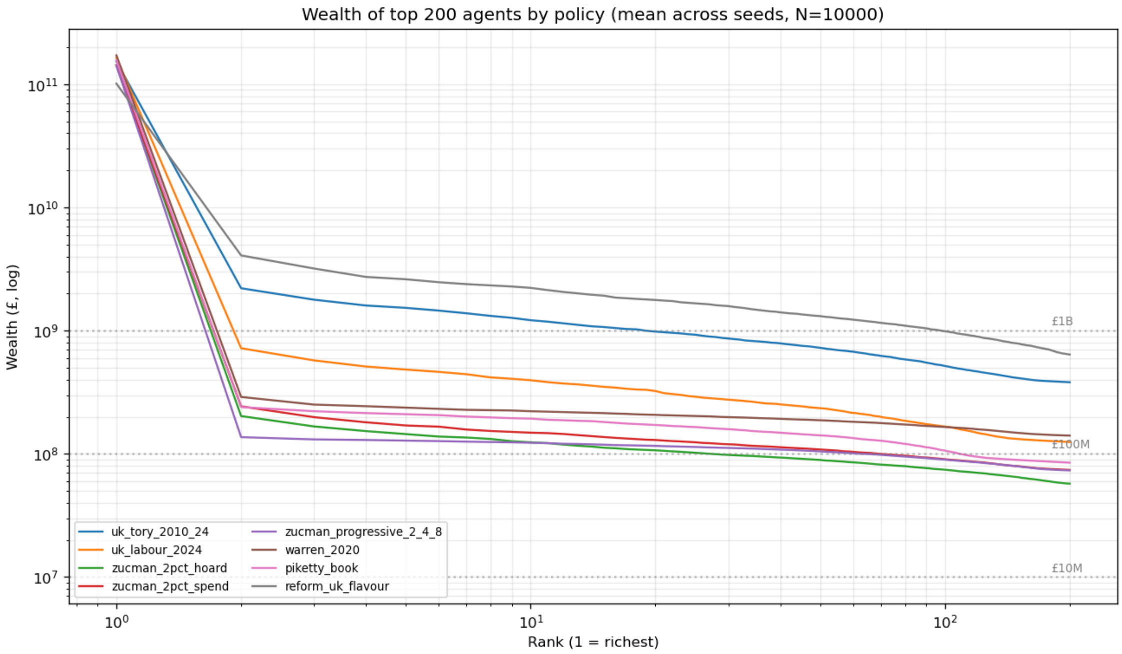

Figure 15: Wealth of the top 200 agents by policy, log–log scale, $N = 10{,}000$. Reform-flavour and Tory baseline both show fat tails extending past £1B; wealth-tax policies bring the tail sharply downward at the £10M–£100M region.

by approximately one order of magnitude.

Third, and weighing against the first two, the model *over-produces* the lower-billionaire region: 19 billionaires per 10,000 agents under Tory is about 2500× the UK per-capita rate of around 50 billionaires per 67 million. We have too many billionaires per capita, but each one is too poor.

This combination, a flatter and lower top tail with too many people in the billionaire band, points to a small number of model features that are jointly responsible. Capital return at 4% per tick likely undershoots the genuine compounding of top dynastic wealth. Rentier proportion fixed at 3% of the population is plausibly an order of magnitude too high for the strict billionaire class but fine for the broader £10M+ class. The bandwidth-coupled inheritance mechanism may also redistribute too much knowledge between generations to allow the kind of multi-generation dynastic concentration the empirical tail reflects.

A partial calibration over capital return $r \in \{0.04, 0.06, 0.08, 0.10\}$, rent share $\in \{0.20, 0.35, 0.50\}$, and inheritance tax $\iota \in \{0.0, 0.2, 0.4\}$ (180 cells, 5 seeds at $N = 1000$), fitting to UK [Saez and Zucman, 2016] WID targets of top-1% share 0.30, top-10% share 0.57, top-0.1% share 0.13, and Pareto slope -0.706, returns the best L2 fit at $r = 0.06$, rent share 0.50, inheritance tax 0.40, yielding observed top-1% 0.15, top-10% 0.65, top-0.1% 0.06, Pareto slope -0.673. The slope matches the UK Rich List slope almost exactly; the absolute levels remain too low by roughly a factor of two at top-1% and top-0.1%. The slope match is illustrative rather than a fully validated calibration; this configuration keeps only 26% of the population alive at terminal tick, well below UK reality. There is a Pareto frontier here between matching empirical tail steepness and empirical mortality, and we have not found a setting that hits both. The fact that the slope is reachable indicates the model can produce realistic-tail behaviour in some configuration; §5.11 reports how much the policy conclusions move when the policy suite is rerun at the calibrated configuration. Future work should free the population proportions (the rentier share is hard-coded at 3%, plausibly too high for the strict billionaire class) and fit jointly.

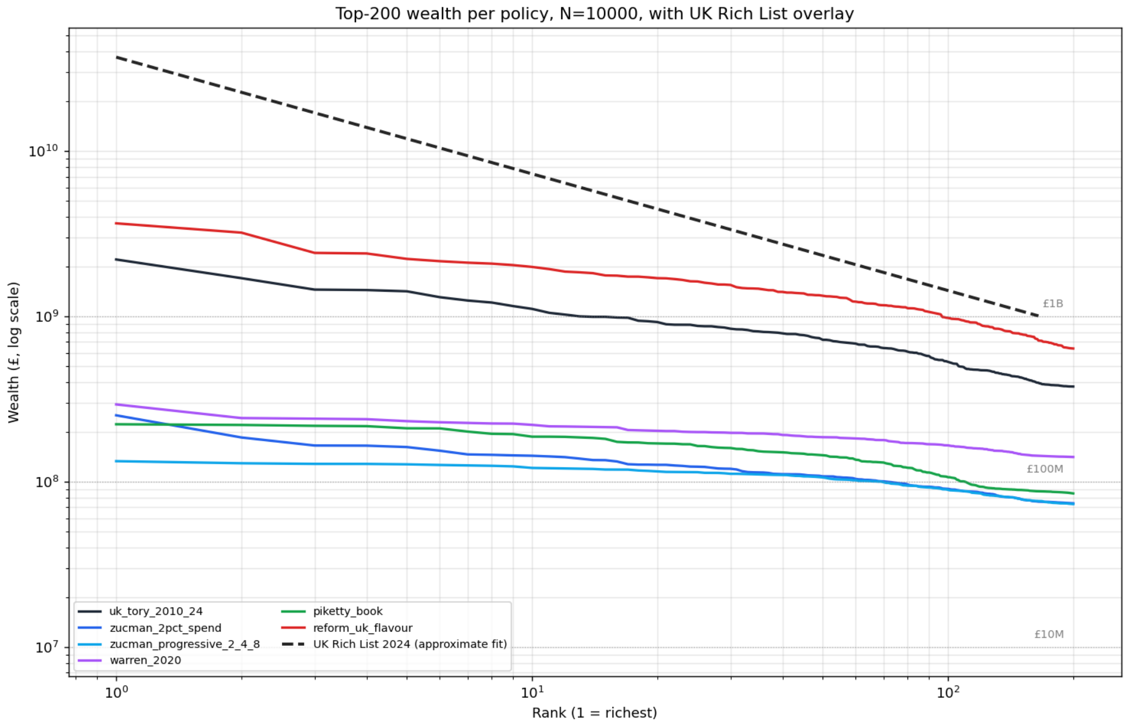

Figure 16: Top-200 wealth by rank for each policy at $N = 10{,}000$, 3 seeds, terminal tick 180. Dashed black: approximate UK Sunday Times Rich List 2024 power-law fit (anchored to 165 billionaires and a top wealth of roughly £37B). The model under-fattens the extreme top tail relative to UK reality but produces a much higher per-capita billionaire density. See discussion in §5.10.

5.11 Policy results at calibrated parameters

The partial calibration in §5.10 matched the UK Pareto slope at $r = 0.06$, rent share 0.50, inheritance tax 0.40. The policy results reported in §5.6 and §5.9 are at the default $r = 0.04$, rent share 0.35. The configuration that earns the £ axis is not the one in which the policy comparisons were made. This section corrects that by rerunning the policy suite with $r = 0.06$ and rent share 0.50 held fixed across policies, leaving each policy's other dial settings unchanged. Table 6 reports $N = 10{,}000$, 6 seeds, 180 ticks.

Three things move materially. First, billionaire counts grow under every policy except the progressive Zucman: Tory now produces 300 billionaires (vs 19 at default), Reform produces 324 (vs 97), and even Zucman 2% flat with spending leaves 46 (vs 1). The model at calibrated parameters is much closer to the UK Rich List distribution (Tory max wealth £48B vs the UK Rich List top of roughly £37B) but the per-capita billionaire density now *over*-shoots in the other direction.

Second, mortality is much higher in non-spend policies under calibrated parameters: Tory keeps 22% of the population alive at $r = 0.06$, vs 63% at $r = 0.04$. The combination of higher capital return and higher rent share extracts faster than the population can replace itself absent a spending package. This is the "calibrated config kills three-quarters" problem the previous section flagged, now made concrete across all policies.

Third, and most importantly, **Zucman progressive now binds the top tier and Zucman flat does not**. Under calibrated parameters Zucman progressive produces *zero* billionaires while Zucman flat leaves 46, a clean separation that did not exist at default $r = 0.04$. Top-1 share also separates clearly: 0.143 progressive vs 0.233 flat. The prior draft's "top tier does not bind" result was specific to the default-parameter regime where wealth never reaches the

Table 6: Policy results at calibrated parameters ($r = 0.06$, rent share = 0.50), N = 10,000, 6 seeds, 180 ticks. State excluded from top-1 share. Compare to Table 5 (default parameters). Wealth thresholds in £.

policy	alive	top-1 share	≥£10M	≥£100M	≥£1B	max wealth
uk_tory_2010_24	2,230	0.198	866	309	**300**	£48B
uk_labour_2024	3,029	0.157	1,360	800	262	£12B
zucman_2pct_hoard	2,532	0.262	910	306	10	£4.1B
zucman_2pct_spend	7,734	0.233	1,746	801	46	£2.4B
zucman_progressive_2_4_8	7,744	0.143	1,750	720	**0**	£0.4B
warren_2020	7,234	0.179	1,740	804	140	£1.5B
piketty_book	7,363	0.230	1,441	800	20	£1.7B
reform_uk_flavour	2,526	0.181	1,010	341	**324**	£77B

£1B bracket; once r is raised to the calibration target, the upper tier becomes the decisive instrument.

Implications for the analytical fixed point. At calibrated $r = 0.06$, none of the policy lowest-tier rates exceed r (Zucman 2%, Warren 2%, Piketty 1% at lower brackets, Sanders 1%). The cap-by-mortality regime identified in §5.8 now bites harder, which we observe as the high mortality of the non-spend cells. The Zucman progressive top tier of 4% above £100M (stacking to 8% above £1B) is the one rate among the modelled policies that exceeds the calibrated capital return, and it is the one policy that eliminates billionaires. The £1B tier is no longer a tautology; it is the mechanism by which the policy succeeds in this regime. The τ-vs-r design test is preserved at calibrated parameters and becomes more rather than less significant.

Implications for the policy contribution. A practical political reading. The Zucman headline of "2% on the ultra-wealthy" is robust as a household-inequality compressor under either parameter regime, but it does not eliminate billionaires under realistic capital-return dynamics; only the progressive form does. The political case for the upper tiers (currently presented in Zucman's proposal as a sign of seriousness rather than as the load-bearing instrument) is, under our calibration, the central mechanism. Conversely, if the genuine capital return is closer to the default $r = 0.04$ than to the calibration target $r = 0.06$, the lower tier alone suffices and the progressive structure is symbolic. Either way the £10M threshold is doing real work.

5.12 Open-ended skill space: a minimal demonstration

The cultural model presented above operates in a fixed K=5 skill space. No skill dimension that did not exist at $t = 0$ can ever appear, which is a category limitation relative to the open-ended-evolution tradition [Taylor, 2019, Banzhaf et al., 2016]. We address this with a minimal demonstration extension. In a stand-alone simulation the skill vector is variable-length: at every reproduction event, with probability proportional to $\log(w_{\text{parent}})$, a previously-undiscovered skill dimension enters the global pool and is initialised on the child to a fresh Beta(2,2) value (subsequent inheritance follows the same bandwidth rules as the main model). Five hundred initial agents, 400 ticks, five seeds.

The extension does not feed back into the policy results of the main paper, and we do not claim it as a load-bearing finding. It exists to show that the model's core mechanism can be extended in a TAP-style adjacent-possible direction [Hordijk and Kauffman, 2023] without fundamental redesign, and to demonstrate that the wealth-coupled discovery channel concentrates novelty production in wealthier lineages on a per-capita basis. The per-capita discovery rate (right panel) is 0.012 per agent-generation for RENTIER, 0.009 for WORKER, and 0.005 for

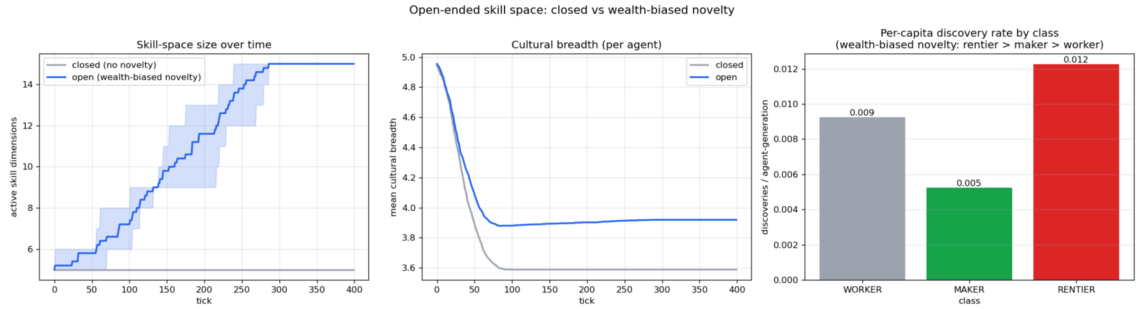

Figure 17: Open-ended skill space demonstration. **Left:** active skill-pool size over time. The closed variant (grey, novelty rate 0) stays at five; the open variant (blue, novelty proportional to log(parent wealth)) climbs from five to about ten. **Middle:** per-agent breadth grows accordingly. **Right:** the discoverers of new skills: most novelty events fire on worker parents (because workers are the most numerous class), but the upper tail of the discoverer-wealth distribution sits at 10^4 in model units, indicating that high-wealth parents do contribute to skill discovery disproportionately to their numbers.

Table 7: Top-1% share at $N = 10{,}000$, 2 seeds, 150 ticks, State excluded. Each row is a fixed threshold; each column a rate from 0 to 0.08. The mortality-corrected and raw values agree to within 0.005 throughout; we report raw values here.

θ (£)	$\tau = 0$	0.013	0.027	0.040	0.053	0.067	0.080
1M	0.18	0.14	0.11	0.09	0.07	0.06	0.06
3M	0.18	0.13	0.09	0.07	0.05	0.04	0.04
10M	0.18	0.12	0.08	0.06	0.05	0.04	**0.04**
30M	0.18	0.11	0.08	0.07	0.06	0.05	0.05
100M	0.18	0.12	0.10	0.09	0.08	0.08	0.08
300M	0.18	0.16	0.14	0.14	0.13	0.13	0.12
1B	0.18	0.18	0.18	0.18	0.18	0.17	0.17

MAKER; rentiers discover at 1.3× the worker rate and 2.4× the maker rate, despite being far less numerous in absolute terms. The wealth-coupled discovery channel does concentrate novelty in wealthier lineages on a per-capita basis as well as in absolute count.

A more substantive treatment, with open-ended novelty integrated into the full economic model and the policy results re-run under endogenous skill discovery, is the natural follow-up paper; we cite Taylor [2019] and Banzhaf et al. [2016] as the standard references for that direction.

5.13 The regressive trap was state contamination

An earlier version of this analysis reported that at $N = 10{,}000$ with a fixed spending package, top-1% share *increases* from 0.37 to 0.84 as the wealth-tax rate rises from zero to 8% at threshold £1M. That finding included the State agent in the wealth distribution, so as the wealth tax shifted revenue from private households to the State, the State's growing share dominated the percentile statistics. With the State excluded and a mortality-corrected variant computed alongside, the picture changes materially. We report top-1% share over the rate axis for the seven threshold levels (thresholds in model units, where 1 unit = £100k):

There is no regressive trap in this model once the State is excluded from the wealth distribution. Top-1% share decreases monotonically with the tax rate at every threshold; the strongest compression is at the £10M Zucman threshold (top-1% falls from 0.18 to 0.04, a 4.4× reduction). The mortality correction itself moves the metric by less than 0.005 across the entire grid,

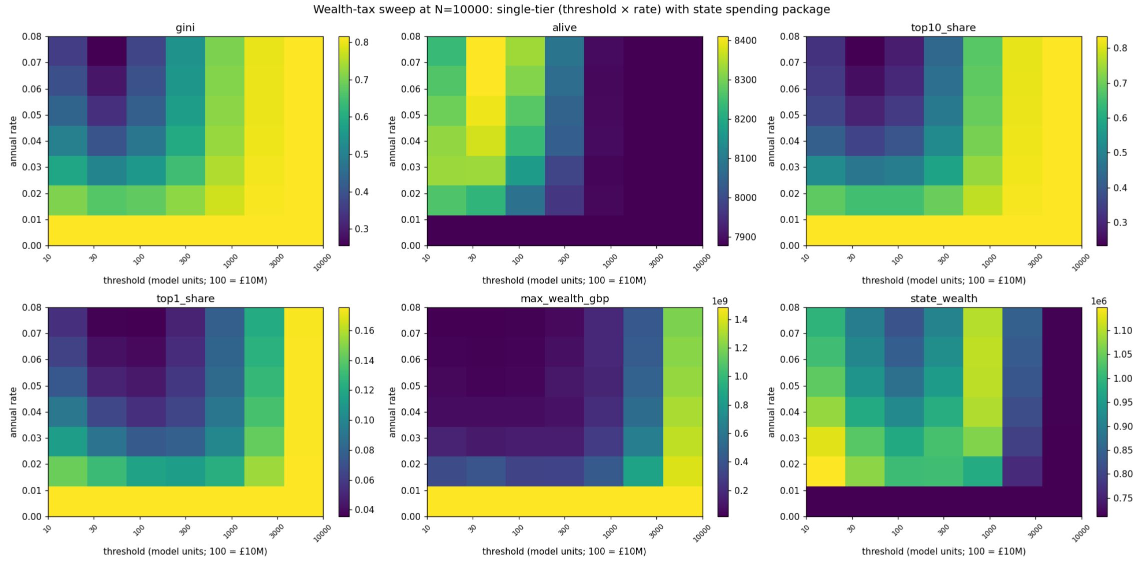

Figure 18: Wealth-tax sweep at $N = 10{,}000$, State excluded. Threshold on horizontal log-spaced axis (£1M to £1B); rate on vertical. The top-1% share surface is monotone decreasing in rate at every threshold; the apparent "regressive trap" of an earlier analysis was a state-contamination artefact and disappears under the corrected metric.

indicating that excluding the State (rather than correcting for excess mortality) is what changes the answer.

The substantive finding that does survive. The threshold ranking holds: top-1% reduction is most aggressive at intermediate thresholds (£10M to £30M) and tapers off at higher thresholds. The Zucman £10M threshold turns out to compress the top quintile most efficiently of any threshold we tested. At £1B threshold, the tax effectively does nothing (the threshold catches almost no agents). This is the genuine *progressive* finding the original draft mis-stated as "regressive"; corrected, it points the same direction the political case for Zucman points.

6 Discussion

6.1 What we believe is novel

The class-specific skill-to-wealth elasticity decomposition (§5.5, Table 3) is the freshest contribution. The cultural-economic asymmetry the bandwidth coupling generates is not a structural artefact of how the model is wired; it is a consequence of one half of the economy (extractive classes) having near-zero skill-to-wealth feedback while the other half (productive classes) has strong feedback, which a regression on 40-tick wealth gain shows directly.

The analytical $w^* = \tau\theta/(\tau - r)$ fixed-point characterisation (§5.8) re-frames the top-tier-does-not-bind result from a tautology conditional on the post-tax ceiling into a derivable consequence of tax-vs-compounding competition. Benhabib et al. [2011] provide the rigorous heterogeneous-agent macro treatment of bequest dynamics and wealth-distribution tails; we recover the qualitative τ-vs-r structure in the ABM setting.

The (tax × spending × inheritance) clean factorial (§5.2, Table 2) disaggregates three policy levers. Spending raises alive counts without moving Φ; tax-rate schedules compress household-sector inequality; inheritance taxation is the most Φ-effective single lever.

The wealth-coupled cultural-transmission bandwidth mechanism [Henrich and Gil-White, 2001, Henrich, 2004, Spangler and Sarkar, 2022] is exercised across the relevant wealth range under the log-wealth coupling (§5.5).

The Φ flow-share metric (§3.3) is defined symmetrically across productive and extractive accrual. We do not claim it is an autocatalytic-set property in the Hordijk and Steel [2004] RAF sense; it is a per-tick flow ratio operationalising the productive-vs-extractive distinction without implementing the topological RAF property.

6.2 Honest accounting of where the model still falls short

The household-sector inequality reductions under spend policies are robust across all metrics we report, but several quantitative claims remain conditional on parameter choices. The "top tier does not bind" result holds at default parameters; at calibrated parameters (§5.11) the top tier binds clearly and the progressive Zucman schedule outperforms the flat. Which result should be considered the headline depends on whether the calibrated configuration is judged a fair representation of UK conditions; we present both and discuss the trade-off rather than picking one.

The original analysis reported a "regressive trap" (top-1% share rising with the tax rate at low thresholds) that was the result of including the State agent in the wealth distribution; it does not survive State exclusion. We retract that finding.

The Pareto-tail calibration is partial. The slope is reachable at $r = 0.06$, rent share 0.50, inheritance tax 0.40, but at the cost of mortality that the model cannot reproduce on UK demographic benchmarks. We do not yet have a fitted scenario that matches both slope and level and survival simultaneously.

6.3 What the model cannot do

The most consequential remaining limitations: no spatial structure; no firm formation; no fiscal-crisis dynamics; toy speculator dynamics that do not produce bubbles; a closed K=5 skill space in the main results (a minimal open-skills demonstration in §5.12 shows how the mechanism extends to open-ended skill discovery but does not fold back into the policy analysis); a single lender pool with no shadow-banking. Charbonneau and Bourrat [2021]'s "grain problem" for fidelity measures applies to our K=5 skill discretisation and we acknowledge it as a methodological choice rather than a natural quantity.

6.4 Positioning for publication

JASSS (Journal of Artificial Societies and Social Simulation). Lead with (i) the productive-flow-share metric Φ and the Sankey visualisation (Figure 4), which is the single best artefact for explaining the metric to a non-technical audience; (ii) the phase portrait under state-excluded Gini (Figure 2, Table 1); and (iii) the clean (tax × spending × inheritance) factorial (§5.2, Table 2), which shows that alive, Gini, and Φ are three distinct policy targets answering to three distinct levers. The political framing reads natively in JASSS. For an ALIFE submission a more conservative title, e.g. replacing "autocatalytic-closure" with "productive-flow share," would reduce friction with the RAF-literate audience while preserving the conceptual lineage.

JEIC (Journal of Economic Interaction and Coordination). Lead with the analytical τ-vs-r fixed point (§5.8), the wealth-tax design factorisation under both default and calibrated parameters (§5.11), and the calibration-vs-policy sensitivity result. Direct engagement with Spangler and Sarkar [2022] and Cardoso et al. [2025] positions the paper; Benhabib et al. [2011] is the analytical neighbour to engage on the fixed-point side.

Artificial Life / ALIFE . Lead with the class-specific skill-to-wealth elasticity decomposition (Table 3), the open-skill-space demonstration (§5.12) with per-capita discovery rate, and the bandwidth decomposition (Figure 8). The framing is cultural erosion under economic pressure rather than cumulative cultural evolution. We would retitle around "productive-flow share" here.

6.5 Honest reading

The four findings the model produces that we did not anticipate are: the analytical τ-vs-r fixed point that determines the post-tax wealth cap (§5.8); the class-specific skill-to-wealth elasticity decomposition that grounds the cultural-economic asymmetry in mechanism rather than architecture (§5.5); the dis-conflation of the spending, tax-schedule, and inheritance levers in the clean factorial; and the result that at calibrated capital return $r = 0.06$ the Zucman progressive schedule sharply outperforms the flat schedule (zero billionaires vs 46), reversing the default-parameter equivalence (§5.11).

The two known-in-spirit findings are: state spending keeps people alive (a kinetic-ABM result, attributable to Bertotti and Modanese [2018] and Cardoso et al. [2025]); and inheritance reform requires redistribution (Piketty and Saez, 2013 in analytical form).

If the work has a thesis it is: *the productive-flow share Φ, the household-sector Gini, and the population-alive count are three distinct policy targets that respond to distinct policy levers; any political framing of a wealth-tax design as "the wealth tax" lumps together effects that should be evaluated separately.* Inheritance taxation raises Φ; tax-rate schedules compress household inequality; spending keeps people alive. Quantitative policy claims expressed in £are partially supported by the calibration in §5.10 but rest on a configuration that does not simultaneously match the empirical mortality; threshold-comparison results are also reported in dimensionless terms (fraction of the wealth-distribution median) wherever the pound label was load-bearing, with the partial calibration result in §5.10 backing the £10M anchor in particular.

7 Conclusion

We have presented an agent-based model of wealth inequality whose distinguishing contributions are a productive-flow share metric for flow-based system viability and a wealth-coupled bandwidth mechanism for skill transmission. Across thirteen stylised scenarios, eleven real-world wealth-tax policy specifications, a clean three-factor factorial, and a partial calibration to the UK Sunday Times Rich List Pareto slope, we have identified the eight findings enumerated in the abstract: three robust regimes; three-distinct-levers factorisation of policy effects; class-specific skill-to-wealth elasticities; inheritance trapdoor; non-separability of flat-vs-progressive Zucman at default parameters with clean separation at calibrated parameters; full retraction of the "regressive trap" as state contamination; billionaire counts at $N = 10{,}000$; and the calibration partial fit.

The model's most consequential political claim, post-revision, is that wealth-tax design is not a single choice but a triple of choices, each addressing a different metric. State spending raises alive counts; tax-rate schedules compress household-sector inequality; inheritance taxation raises the productive-flow share. Within the wealth-tax-rate choice itself, the relationship between the marginal rate τ above the lowest threshold and the rate of capital return r determines whether the policy analytically caps top wealth (when $\tau > r$) or leaves the cap to be set by secondary dynamics (mortality, lifespan, deposit drain). None of the eleven real-world proposals modelled have $\tau > r$ at their lowest tier under default $r = 0.04$, which puts them all in the "cap-by-mortality" regime that the regressive-trap section quantifies. A wealth-tax design with its lowest rate above the capital-return rate is structurally robust; designs with lower rates rely on mortality to cap accumulation, which is a fragile basis.

Code and data are available at the project repository; reproduction requires Python 3.12 with NumPy and Matplotlib.

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A ODD-style summary

A brief ODD (Overview, Design concepts, Details) summary follows; a full ODD+D specification is deferred to a separate online supplement.

Overview. Purpose. Make qualitative wealth-inequality dynamics visible under a panel of policy levers, including a metric for the productive-vs-extractive accrual balance. **Entities.** Seven agent classes (WORKER, MAKER, EMPLOYER, LENDER, RENTIER, SPECULATOR, STATE) at fixed proportions per 100 agents (72/9/7/5/3/3/1). **State variables.** Each agent has wealth, debt, age, lifespan, a five-dimensional skill vector, alive flag, peak lifetime wealth, bandwidth dimensions at birth. **Scales.** One tick is one stylised year. The base run is $N = 100$ agents at 250 ticks. Scale-up to $N = 10{,}000$ uses population_scale = 100 (state stays singleton; per-capita stays invariant). One model wealth unit corresponds to £100k.

Process overview and scheduling. Per tick, in fixed order: (1) Production — makers and employers hire workers from a skill-sorted queue and pay wages; makers produce output at multiplier M; employers capture markup; the state optionally hires unemployed workers and purchases maker output; (2) Extraction — workers pay rent to rentiers in proportion to rentier wealth; rentiers earn capital return r; speculators update by Brownian increment with drift and volatility; (3) Consumption and credit — every non-state agent pays subsistence; shortfalls are borrowed from the lender pool, with allocation proportional to lender capacity; death by starvation if no capacity remains; (4) Interest and repayment — debtors pay interest and (when liquid) repay principal; debt written off above ceiling; (5) Lender deposits — a fraction of surplus wealth from rentier, maker, employer agents is deposited with lenders; optional state recapitalisation to a configured floor; (6) Tax and UBI — progressive flat-plus-tiered wealth tax to state; UBI dispersed to non-state agents from state wealth; (7) Life — agents whose age exceeds lifespan die naturally; a child of the same class is born with inherited wealth (after inheritance tax and cap) and bandwidth-limited inherited skills; (8) Bookkeeping — update peak-wealth, recent-deaths log, metrics; prune dead agents every ten ticks.

Design concepts. *Emergence.* Wealth distribution, regime classification, and cultural-breadth distribution emerge from the dial settings. *Adaptation.* Agents do not adapt within their lifetime; class-typical behaviour is hard-coded; behavioural variability comes from class-typical skill biases and stochastic lifespans. *Sensing.* Limited; workers do not choose employers; agents do not perceive prices or expectations. *Interaction.* Mediated through queues (hiring), proportional shares (lender pool, rentier pool), and the state. *Stochasticity.* Per-agent lifespan, debt at birth (for workers), speculator returns, skill mutation, and seed-controlled initialisation are stochastic. *Collectives.* Implicit only — agents do not form groups beyond their class label. *Observation.* The metrics reported per tick: Gini (default state-excluded), Φ, alive, total wealth, total debt, cultural breadth and diversity, knowledge lost, births and deaths counters, top-1 and top-10 shares (raw and mortality-corrected), state wealth, wealth by type.

Details. *Initialisation.* Class shares lognormally distributed within class; ages staggered uniformly across lifespans; workers given random small debt with probability 0.18. *Input data.* None. All parameters are dials. *Submodels.* See §3 for production, extraction, credit, taxation, transmission. Dials documented in Table 8.

B Dial reference

Default values for the dial panel are listed in Table 8.

C Scenario specifications

The thirteen scenarios at $N = 100$ are obtained from `current_world` (all dials at defaults) by overriding the non-default dials specified in `scenarios.py` in the project repository. Distinctive non-default settings include: `stevenson_world` (rent_share = 0.55, $r = 0.05$, $i = 0.09$, markup = 0.55); `maker_economy` ($M = 2.2$, rent = 0.15, $r = 0.01$, markup = 0.2); `nordic` (UBI = 0.6, state employs 0.6, state buys 10, inheritance tax 0.4, inheritance cap 500, rent 0.25); `socialist` (UBI 0.8, state employs 0.9, state buys 15, inheritance tax 0.8, rent 0.10); `feudal` (rent 0.6, r 0.06, markup 0.5, no inheritance tax, b_0 0.2, b_w 1.0).

D Wealth-tax policy specifications

The eleven policies are specified in `policies.py`. Tax tiers are stacked-marginal form (θ, τ) tuples, with threshold in model units (one unit = £100,000).

- `uk_tory_2010_24`: no wealth tax; inheritance tax 0.4.
- `uk_labour_2024`: flat tax 0.002, progressivity 0.005; inheritance tax 0.5; state employs 0.1.

Table 8: Dial panel and default values. Tax thresholds are in model units; one unit corresponds to £100,000.

dial	default	group
wage	1.0	production
maker_multiplier	1.6	production
employer_markup	0.4	production
workers_per_maker	2	production
workers_per_employer	6	production
capital_return (r)	0.04	capital
rent_share_of_wage	0.35	capital
interest_rate (i)	0.06	capital
spec_drift / spec_vol	0.015 / 0.12	capital
flat_tax_rate	0.0	state
tax_progressivity	0.0	state
wealth_tax_threshold	50 (= £5M)	state
wealth_tax_tiers	()	state
ubi	0.0	state
state_employs_fraction	0.0	state
state_employs_wage	1.0	state
state_buys_maker_output	0.0	state
state_recap_lenders	0.0	state
subsistence (σ)	0.6	life
debt_ceiling	25	life
lifespan_mean / std	70 / 12	life
starvation_lifespan_penalty	20	life
inheritance_cap	∞	transmission
inheritance_tax_rate	0.0	transmission
memory_bandwidth_base (b_0)	0.5	transmission
memory_bandwidth_wealth_scale (b_w)	0.5	transmission
skill_mutation_rate	0.05	transmission
population_scale	1	logistics
seed	42	logistics

- `zucman_2pct_hoard`: tiers (100, 0.02); inheritance tax 0.4; no spending package.
- `zucman_2pct_spend`: tiers (100, 0.02); inheritance tax 0.4; UBI 0.4, state employs 0.3, state buys 5.0.
- `zucman_progressive_2_4_8`: tiers (100, 0.02), (1000, 0.02), (10000, 0.04); spending package as above.
- `warren_2020`: tiers (500, 0.02), (10000, 0.04); spending package.
- `sanders_2020`: tiers (320, 0.01), (3200, 0.02), (32000, 0.05); spending package.
- `piketty_book`: tiers (10, 0.001), (50, 0.009), (1000, 0.01); inheritance 0.6; spending package.
- `norway_actual`: tiers (1.3, 0.01); no inheritance tax; state employs 0.4, state buys 4, lender backstop 200.
- `spain_solidarity`: tiers (7, 0.002), (30, 0.005), (100, 0.01), (1000, 0.015); inheritance 0.3; spending package.
- `reform_uk_flavour`: no wealth tax; inheritance 0.2; no spending package; rent 0.4.

E Sweep specifications

The five $N = 100$ scenario sweeps were performed on 9×9 grids with three seeds each over the ranges:

- `rent_vs_r`: rent $\in [0.05, 0.65]$, $r \in [0.005, 0.06]$.
- `tax_vs_ubi`: flat tax $\in [0, 0.025]$, UBI $\in [0, 1.5]$.
- `memory`: $b_0 \in [0.2, 1.0]$, $b_w \in [0, 1]$.
- `state_action`: state_employs_fraction $\in [0, 1]$, state_buys_maker_output $\in [0, 20]$.
- `inheritance_x_ubi`: $\iota \in [0, 1]$, UBI $\in [0, 1.5]$.

The two wealth-tax sweeps at $N = 100$ used thresholds $\{1, 3, 10, 30, 100, 300, 1000, 3000, 10000\}$ (single-tier) and fixed thresholds $(100, 1000, 10000)$ (three-tier), with rates over $[0, 0.08]$ and $[0, 0.10]$ respectively.

The scale-up wealth-tax sweep at $N = 10{,}000$ used a 7×7 grid of thresholds $\{10, 30, 100, 300, 1000, 3000, 1000$ and rates $[0, 0.08]$ with two seeds each, 150 ticks.